

 Weatherford®		DUAL LATEROLOG MICROLATEROLOG	
COMPANY WELL FIELD PROVINCE/COUNTY COUNTRY/STATE LOCATION		S.N.G.N. ROMGAZ S.A. 209 NADES NADES MURES ROMANIA X = 529925.187 m Y = 477271.675 m	
Latitude Longitude		46°16'07.03" N 24°42'12.69" E	
Permanent Datum GL, Elevation 480.21 metres Log Measured From KB, 4.50 metres above Permanent Datum Drilling Measured From KB		Other Services XY CALIPER BOREHOLE NAVIGATION COMPENSATED SONIC CBL PHOTODENSITY DUAL NEUTRON Elevations: KB 484.71 DF 484.71 GL 480.21	
Date		04-SEP-2024	
Run Number		2A+2B	
Service Order		24220368	
Depth Driller		1225.00 metres	
Depth Logger		1226.00 metres	
First Reading		1221.00 metres	
Last Reading		250.00 metres	
Casing Driller		250.00 metres	
Casing Logger		249.50 metres	
Bit Size		8.500 inches	
Hole Fluid Type		INHIBITIV	
Density / Viscosity		1.10 g/cc 50.00 SEC/L	
PH / Fluid Loss		9.50 4.10 ml/30Min	
Sample Source		FLOWLINE	
Rm @ Measured Temp		0.15 @ 25.0 ohm-m	
Rmf @ Measured Temp		0.12 @ 25.0 ohm-m	
Rmc @ Measured Temp		0.18 @ 25.0 ohm-m	
Source Rmf / Rmc		PRESS FILTER	
Rm @ BHT		0.11 @ 39.0 ohm-m	
Time Since Circulation		6 HOURS	
Max Recorded Temp		39.00 deg C	
Equipment / Base		13347 PLO	
Recorded By		MALEA CRISTIAN	
Witnessed By		VASILE FARCAS	

In interpreting, communicating or providing information and/or making recommendations, either written or oral, as to logs or test or other data, type or amount of material, or Work or other service to be furnished, or manner of performance, or in predicting results to be obtained, the Contractor will give the Company the benefit of the Contractor's best judgment based on its experience and will perform all such Work in a good and workmanlike manner. Any interpretation of test or other data, and any recommendation or reservoir description based upon such interpretations, are opinions based upon inferences from measurements and empirical relationships and assumptions, which inferences and assumptions are not infallible, and with respect to which professional engineers and analysts may differ. ACCORDINGLY ANY INTERPRETATION OR RECOMMENDATION RESULTING FROM THE SERVICES WILL BE AT THE SOLE RISK OF THE COMPANY, AND THE CONTRACTOR CANNOT AND DOES NOT WARRANT THE ACCURACY, CORRECTNESS OR COMPLETENESS OF ANY SUCH INTERPRETATION OR RECOMMENDATION, WHICH INTERPRETATIONS AND RECOMMENDATIONS SHOULD NOT, THEREFORE, UNDER ANY CIRCUMSTANCES BE RELIED UPON AS THE SOLE OR MAIN BASIS FOR ANY DRILLING, COMPLETION, WELL TREATMENT, PRODUCTION OR FINANCIAL DECISION, OR ANY PROCEDURE INVOLVING ANY RISK TO THE SAFETY OF ANY DRILLING ACTIVITY, DRILLING RIG OR ITS CREW OR ANY OTHER INDIVIDUAL. THE COMPANY HAS FULL RESPONSIBILITY FOR ALL DECISIONS CONCERNING THE SERVICES.

BOREHOLE RECORD			Last Edited: 04-SEP-2024 10:04	
Bit Size inches	Depth From metres	Depth To metres		
12.250	23.00	246.00		
8.500	250.00	1225.00		

CASING RECORD				
Type	Size inches	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	16.000	0.00	23.00	0.00
CASING	9.625	0.00	250.00	43.50

REMARKS

Clients intervals and scales logged

Acquisition software: Weatherford Well Manager, version 24.03.1279

Cable used: CAMESA size 15/32", type 7J46RXZ with cable head and 8k weakpoint set

Depth wheel calibration factor used 0.6096

Depth Control Procedure for Multiple runs in the well followed.

Run2A: CBHC, MTAK, MBE, MBE, SHA, MCG, MBN, MTC-Y, MSS, MTC-X, SKJ, MUG, MLE, MLG, MTI, MGS,HFS

Run2B: CBHC, MTAK, SHA, MCG, SKJ, MMR, MDN, MPD, HFS

Run2C: CBHC, MTAK, SHA, MCG, MISD, SKJ, MISE, SKJ, MISD, MSS, MISD, HFS logged in combination.

Rm, Rmf, Rmc measured on samples from Mud Engineer given on 04.09.2024

Caliper measurements and averages from TD to casing:

Total hole volume = 35.32 cubic meters

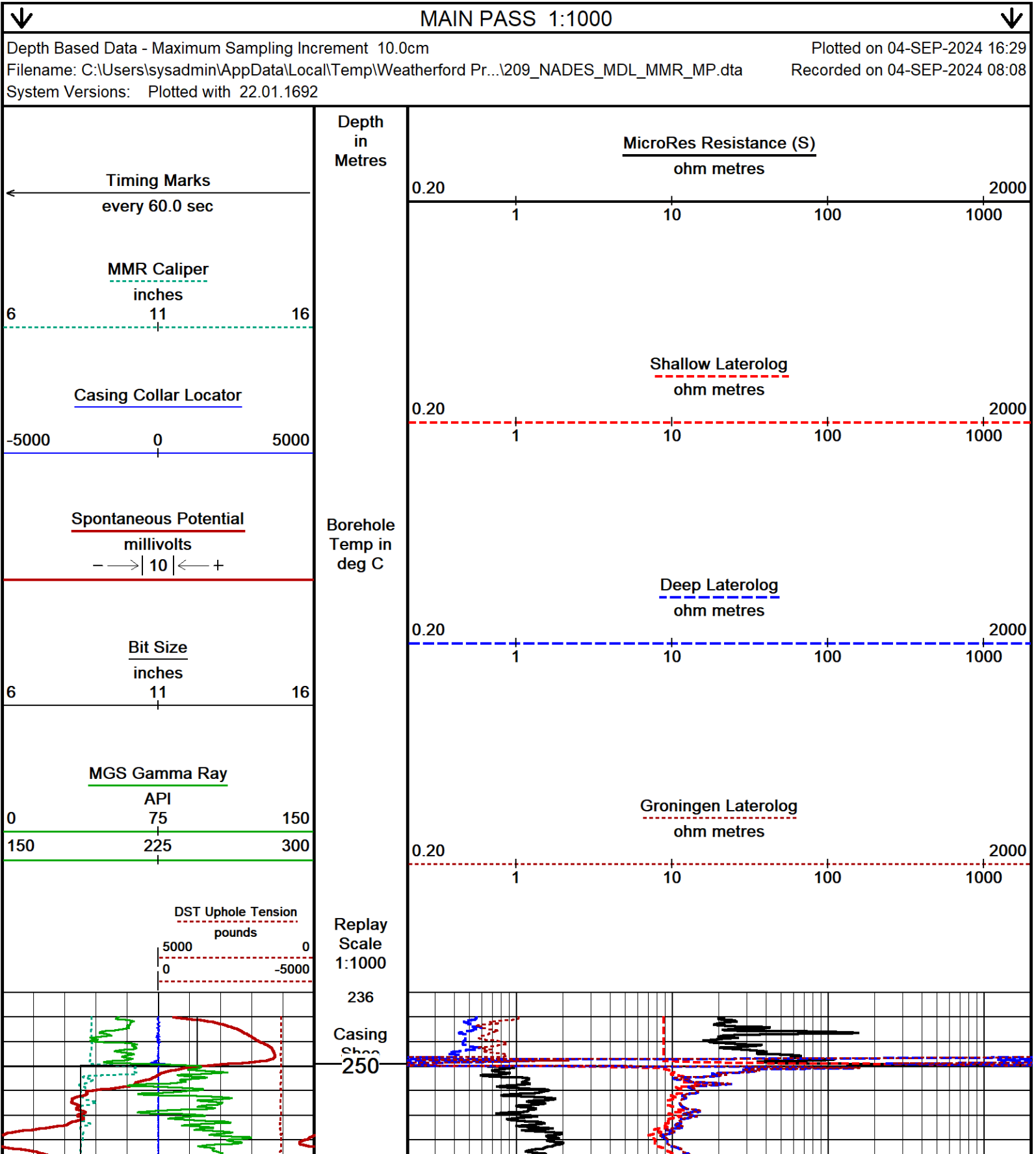
Annular volume = 20.52 cubic meters, based on future casing size of 5.5 inches

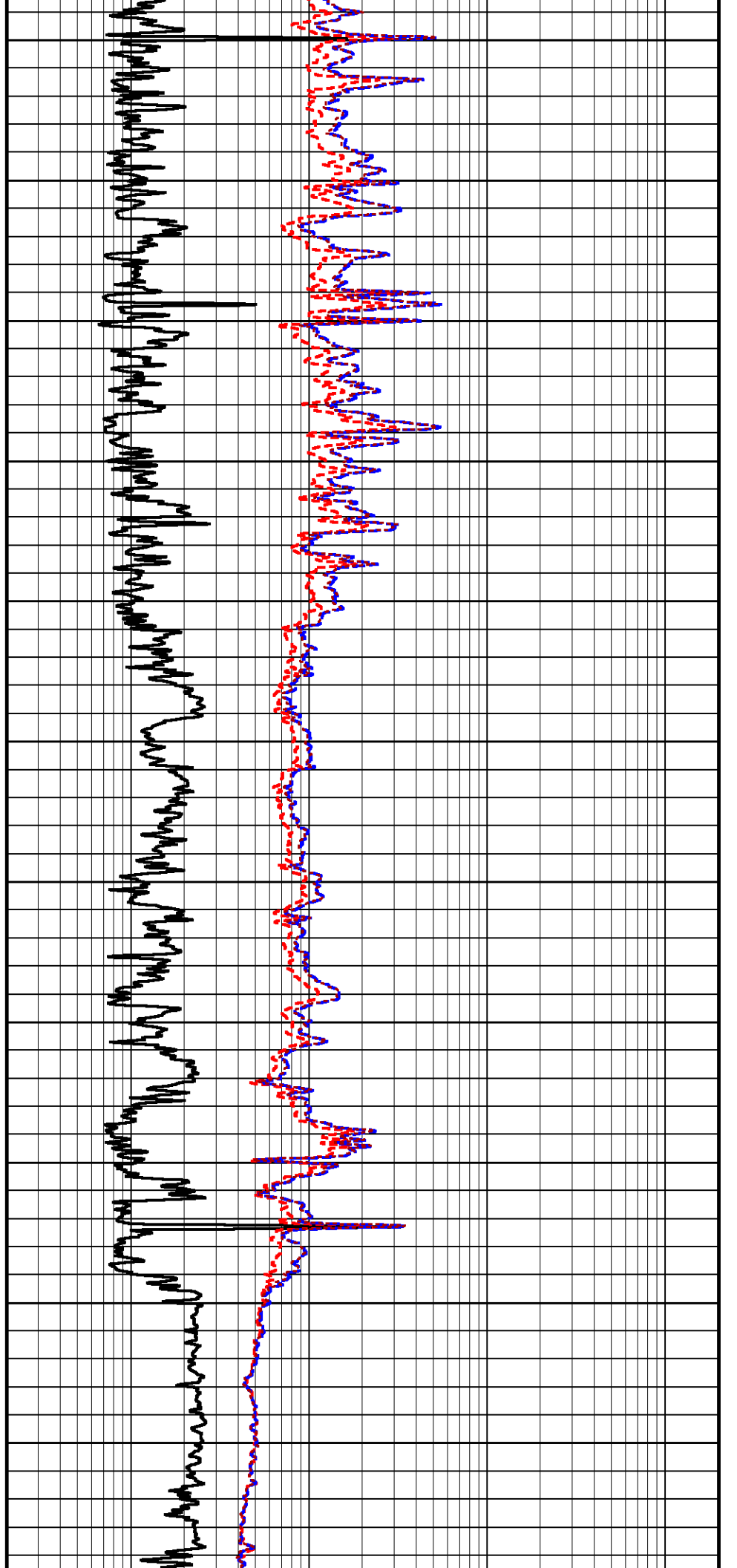
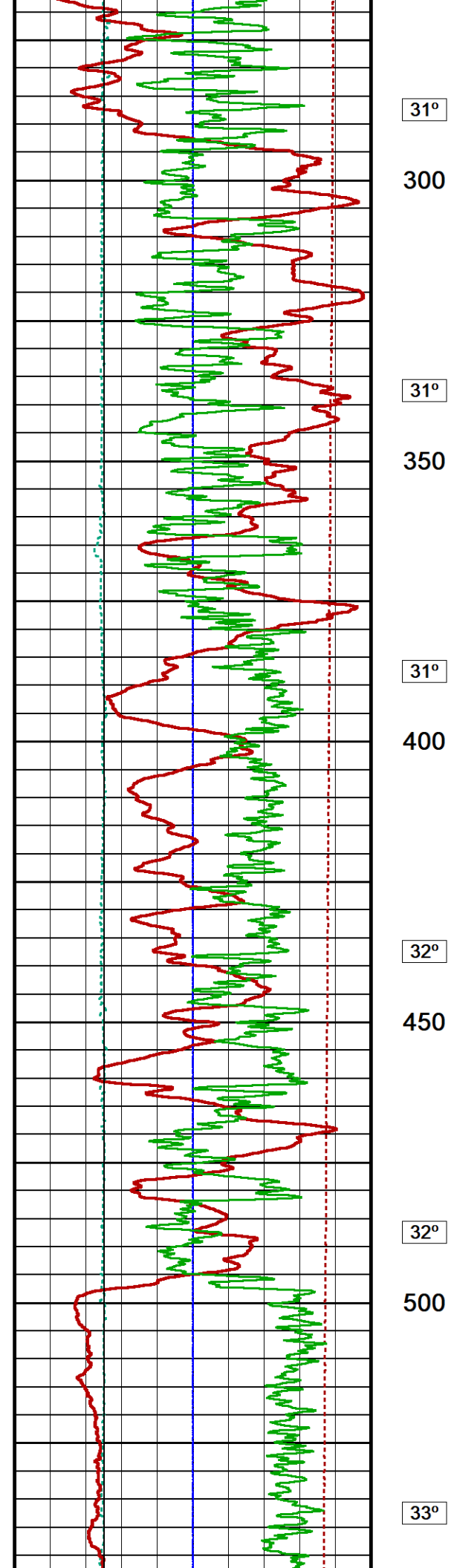
Average hole diameter = 8.49 inches (215.77 mm), based on XY caliper

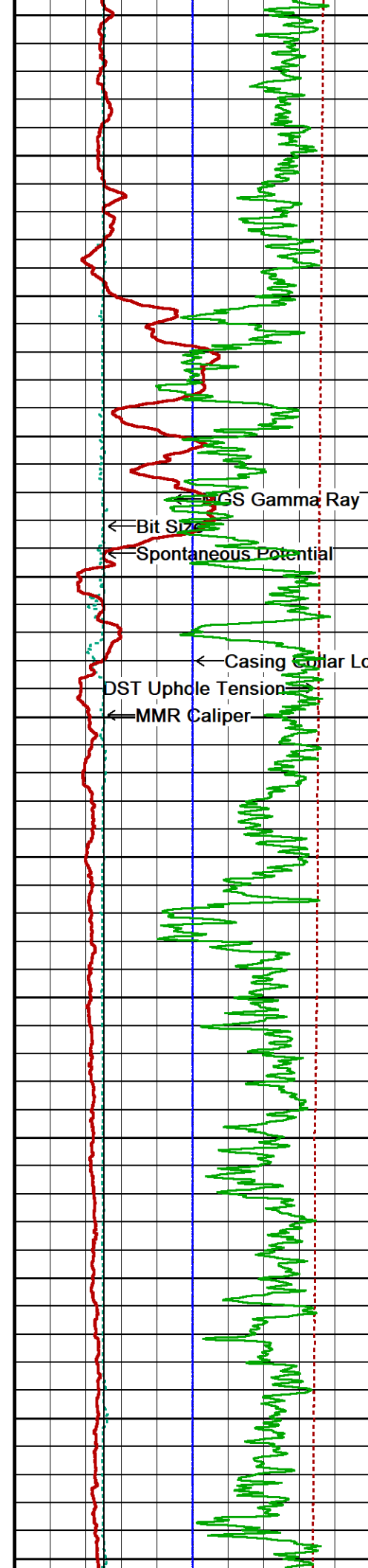
AFTER SURVEY CHECKS			
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView90\0\209_NADES_RES_CALB.dta			
Caliper Check MTC-B.A 123		Field Calibration on 03-SEP-2024 09:37 After Survey Check on 04-SEP-2024 10:05	
	Before (in) 7.96	After (in) 8.81	
Caliper Check Tolerances MTC-B.A 123			
Long Arm After Check	8.81	8.358.759.15	in
Caliper Check MTC-B.A 102		Field Calibration on 03-SEP-2024 09:35 After Survey Check on 04-SEP-2024 10:06	
	Before (in) 7.96	After (in) 8.82	
Caliper Check Tolerances MTC-B.A 102			
Long Arm After Check	8.82	8.358.759.15	in
Laterolog Check MLE-D.A 283		Before Survey Check on 04-SEP-2024 07:22 After Survey Check on 04-SEP-2024 09:56	
Channel	Before Survey (ohm-m)	After Survey (ohm-m)	
Shallow	46.0	46.0	
Deep	28.5	28.5	
Groningen	231.3	231.3	
Laterolog Check Tolerances MLE-D.A 283			
Shallow After Check	46.0	-2%46.0+2%	ohm-m
Deep After Check	28.5	-2%28.5+2%	ohm-m
Groningen After Check	231.3	-2%231.3+2%	ohm-m

AFTER SURVEY CHECKS			
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView90\0\209_NADES_NUC_CALB.dta			
Caliper Check MMR-C.A 219		Field Calibration on 04-SEP-2024 06:09 After Survey Check on 04-SEP-2024 13:33	
	Before (in) 7.96	After (in) 8.86	
Caliper Check Tolerances MMR-C.A 219			
		8.558.758.95	

Short Arm After Check	8.86		in
Micro Laterolog Check MMR-C.A 219		Before Survey Check on 04-SEP-2024 06:10 After Survey Check on 04-SEP-2024 13:35	
	Before Survey (ohm-m)	After Survey (ohm-m)	
	5.1	5.1	
Micro Laterolog Check Tolerances MMR-C.A 219			
After Check	5.1	-2% 5.1 +2%	ohm-m







550

33°

600

34°

650

34°

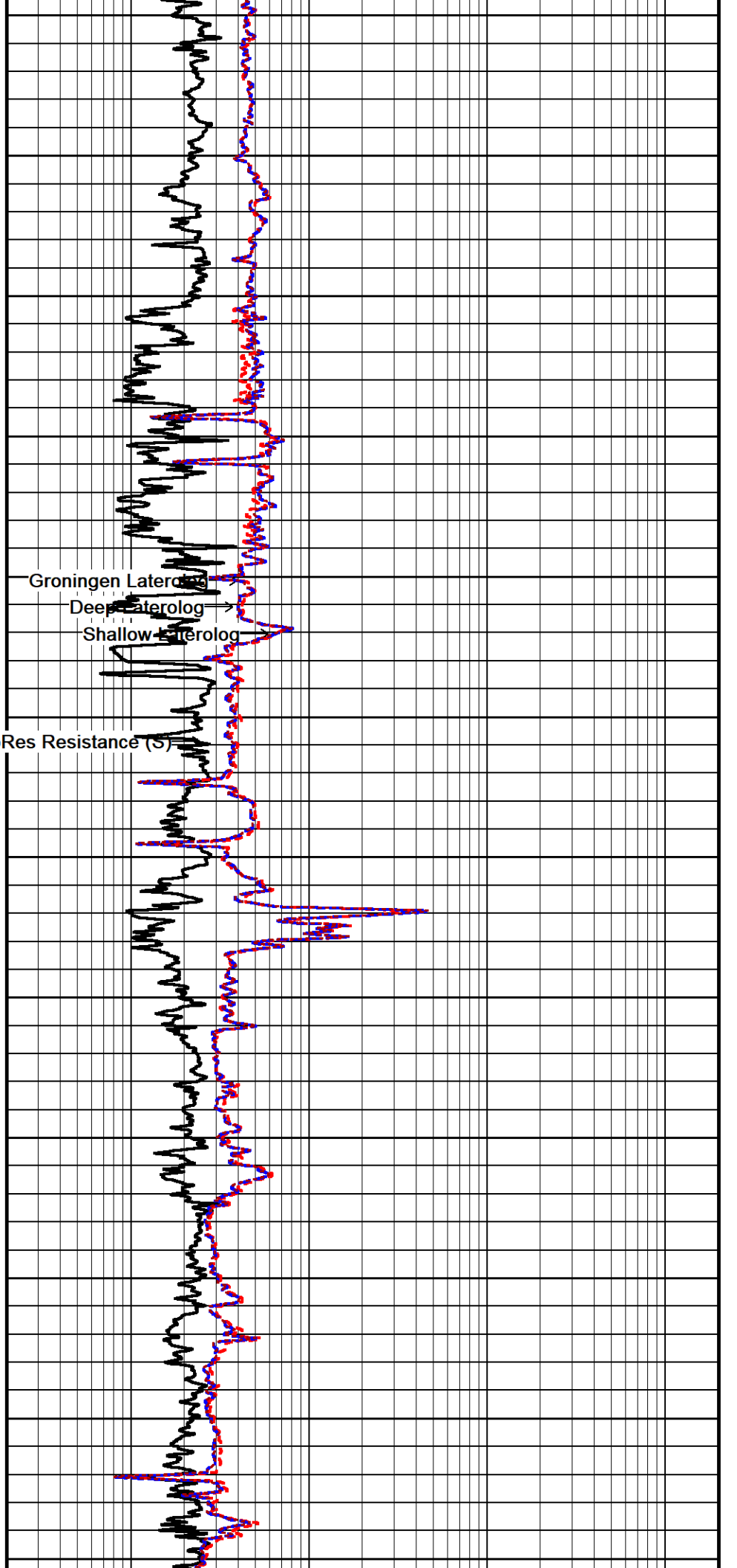
700

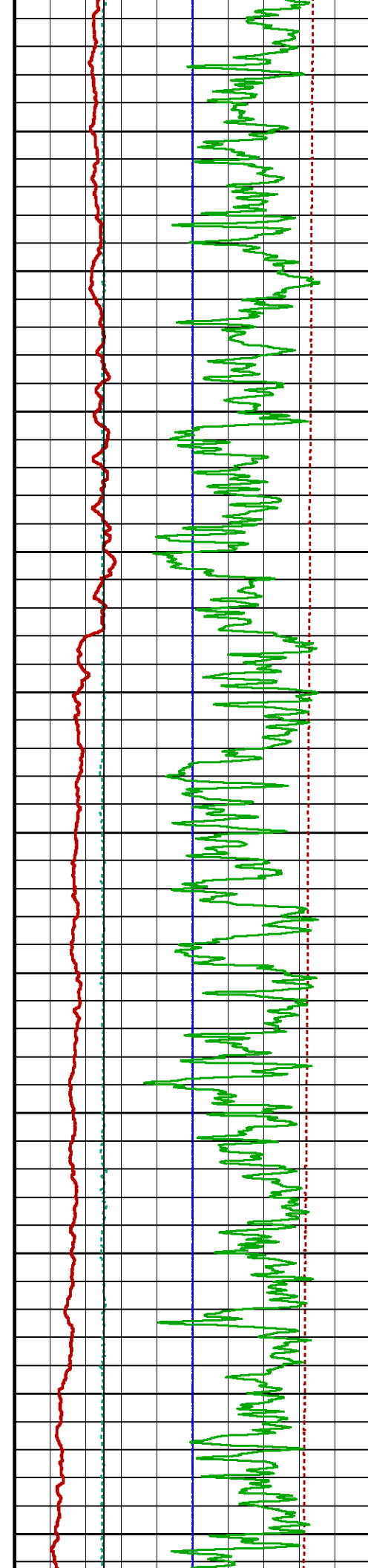
35°

750

35°

800





36°

850

36°

900

37°

950

37°

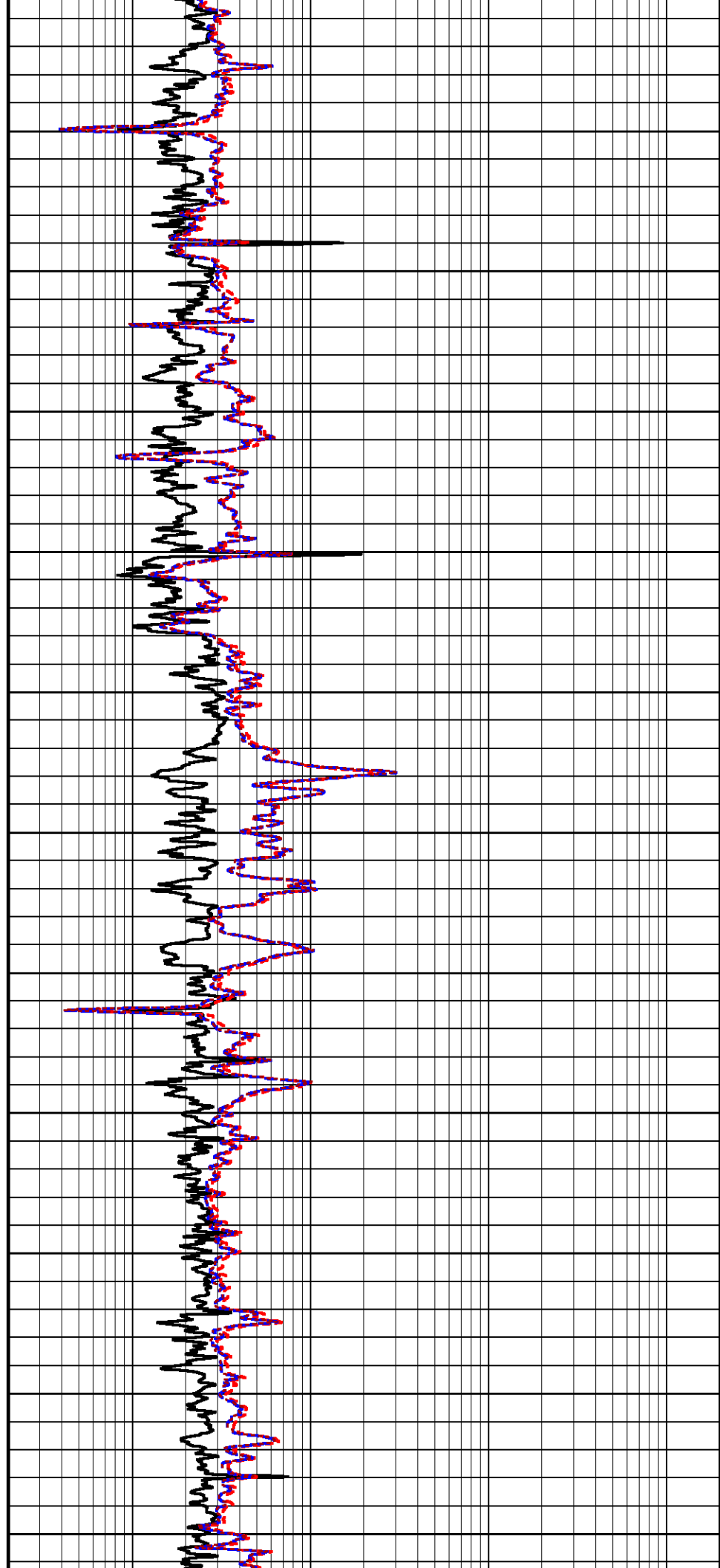
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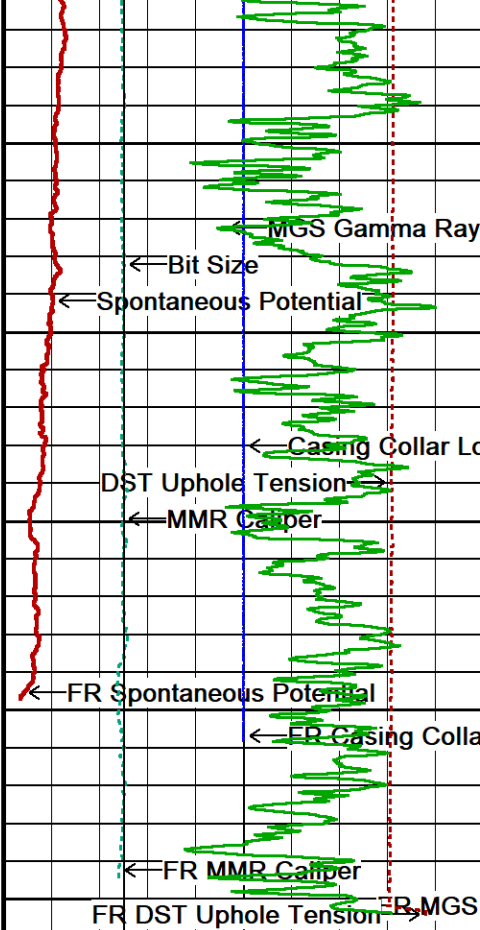
38°

1050

38°

1100





39°

1150

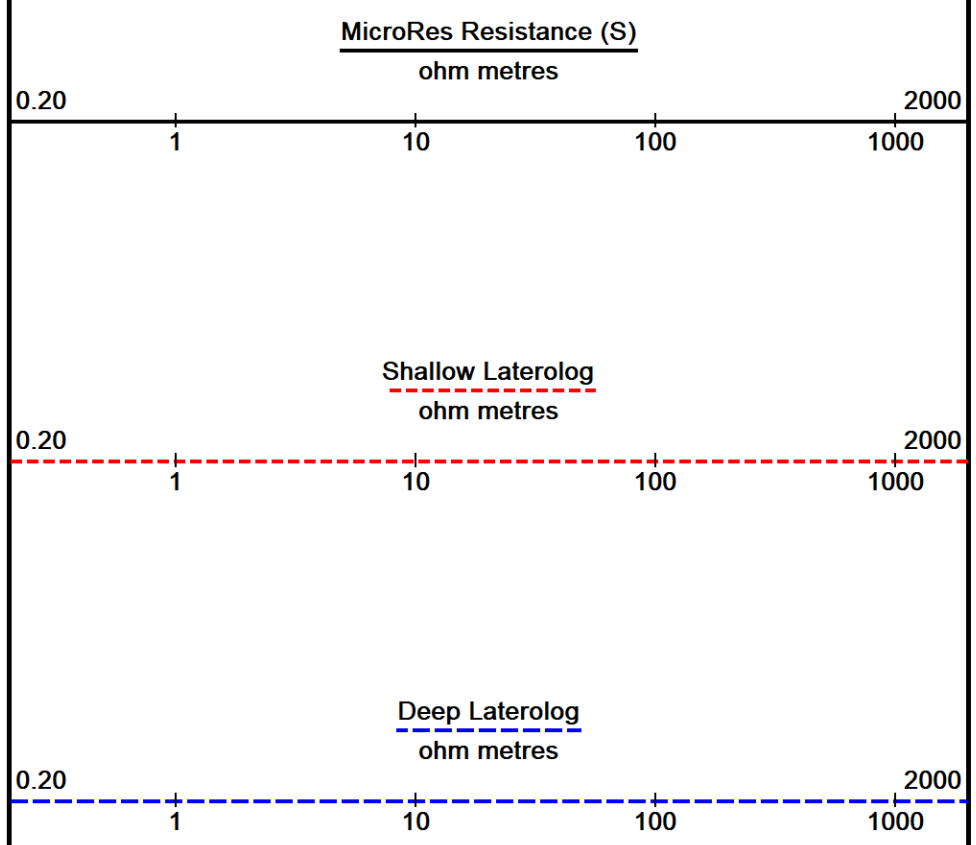
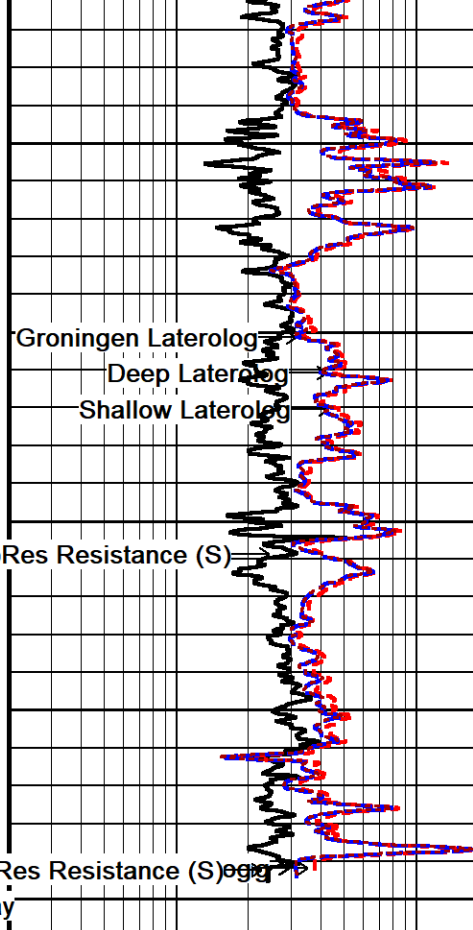
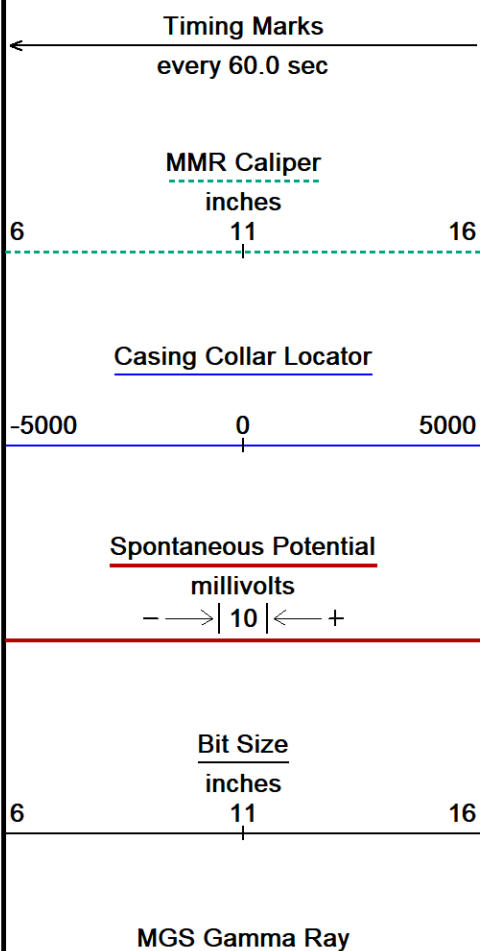
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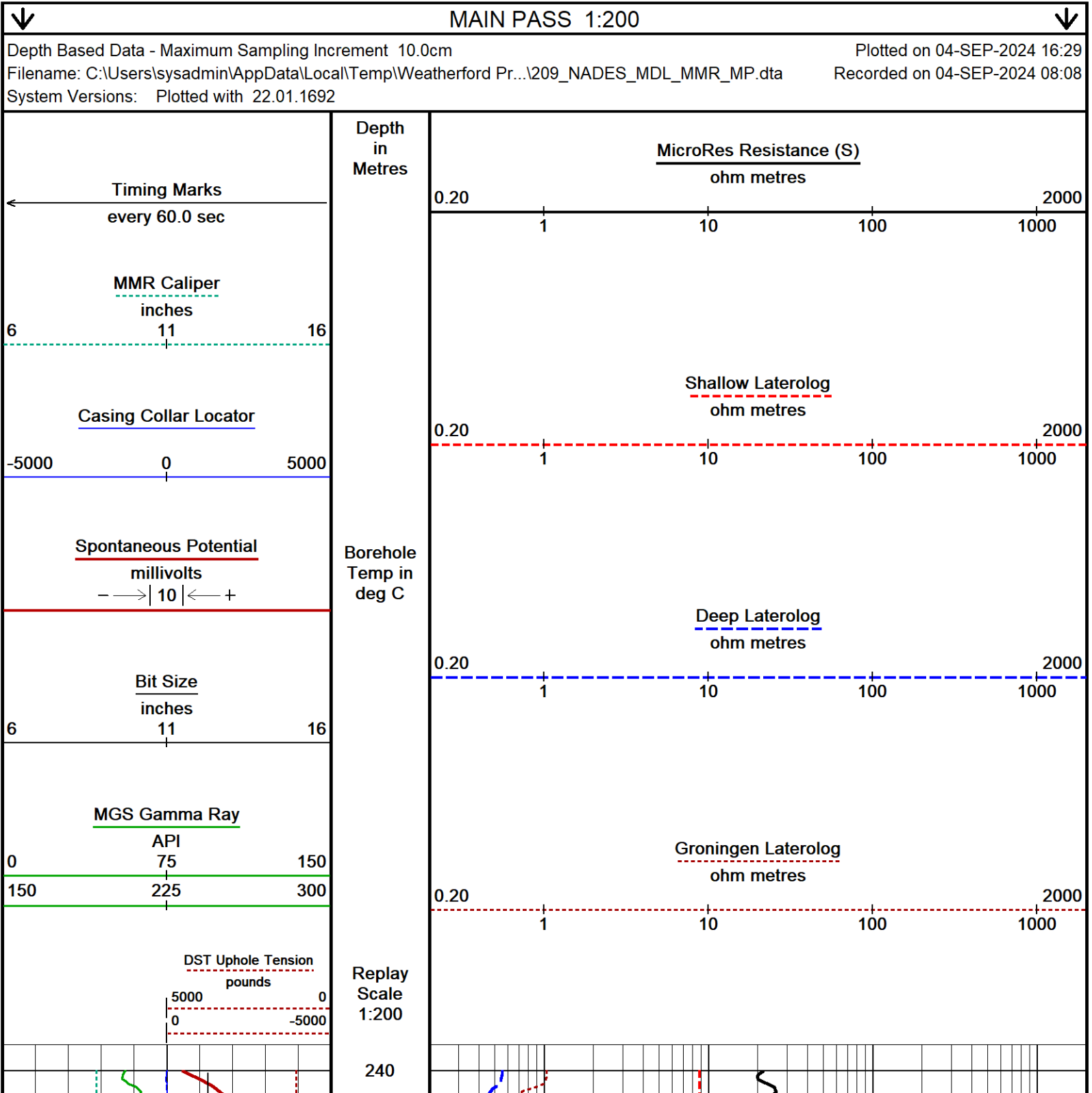
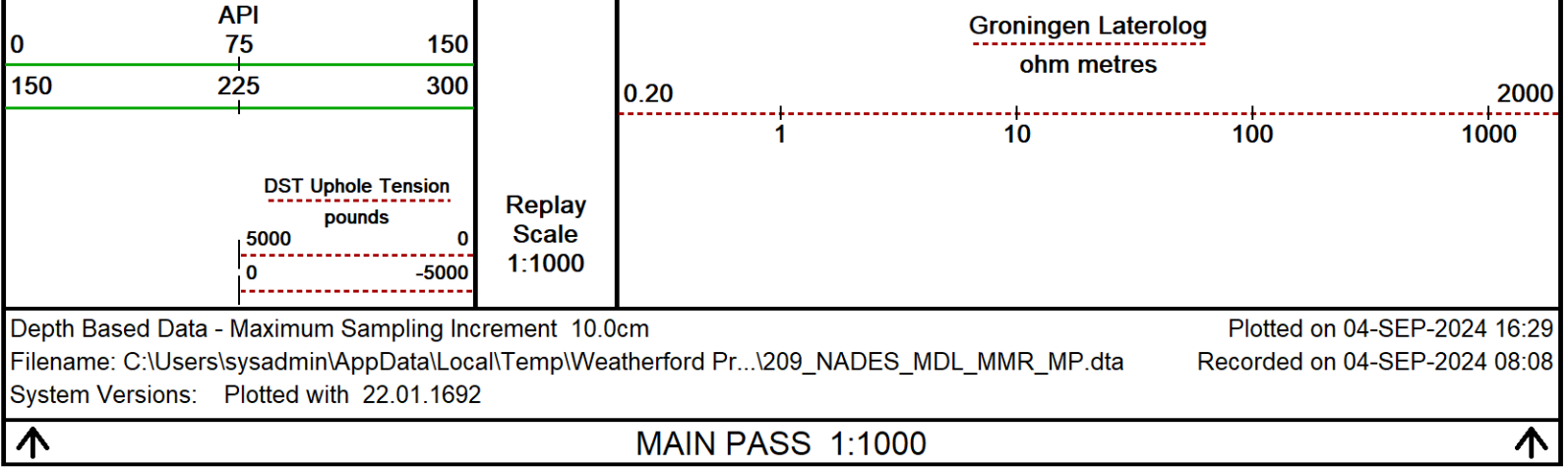
1200

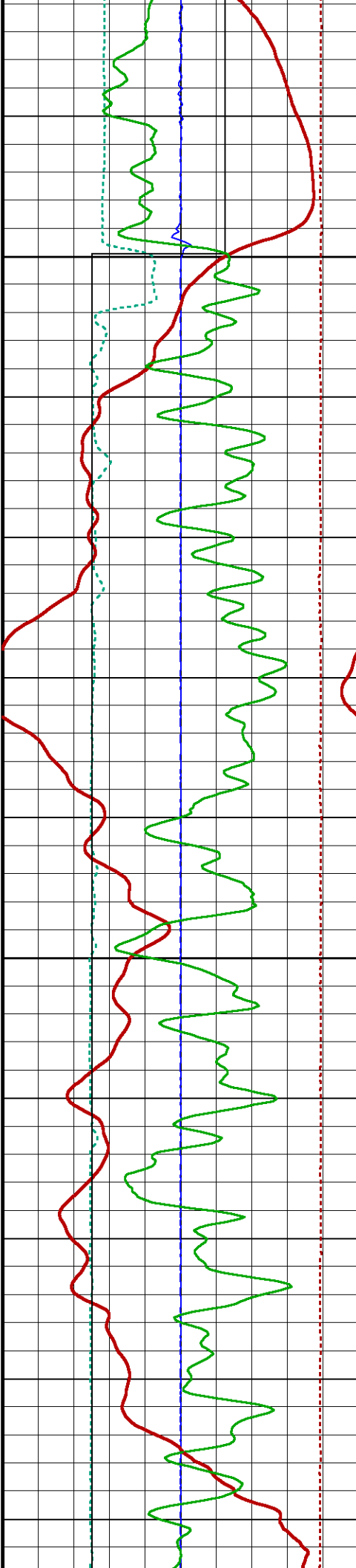
TD

Depth in Metres

Borehole Temp in deg C







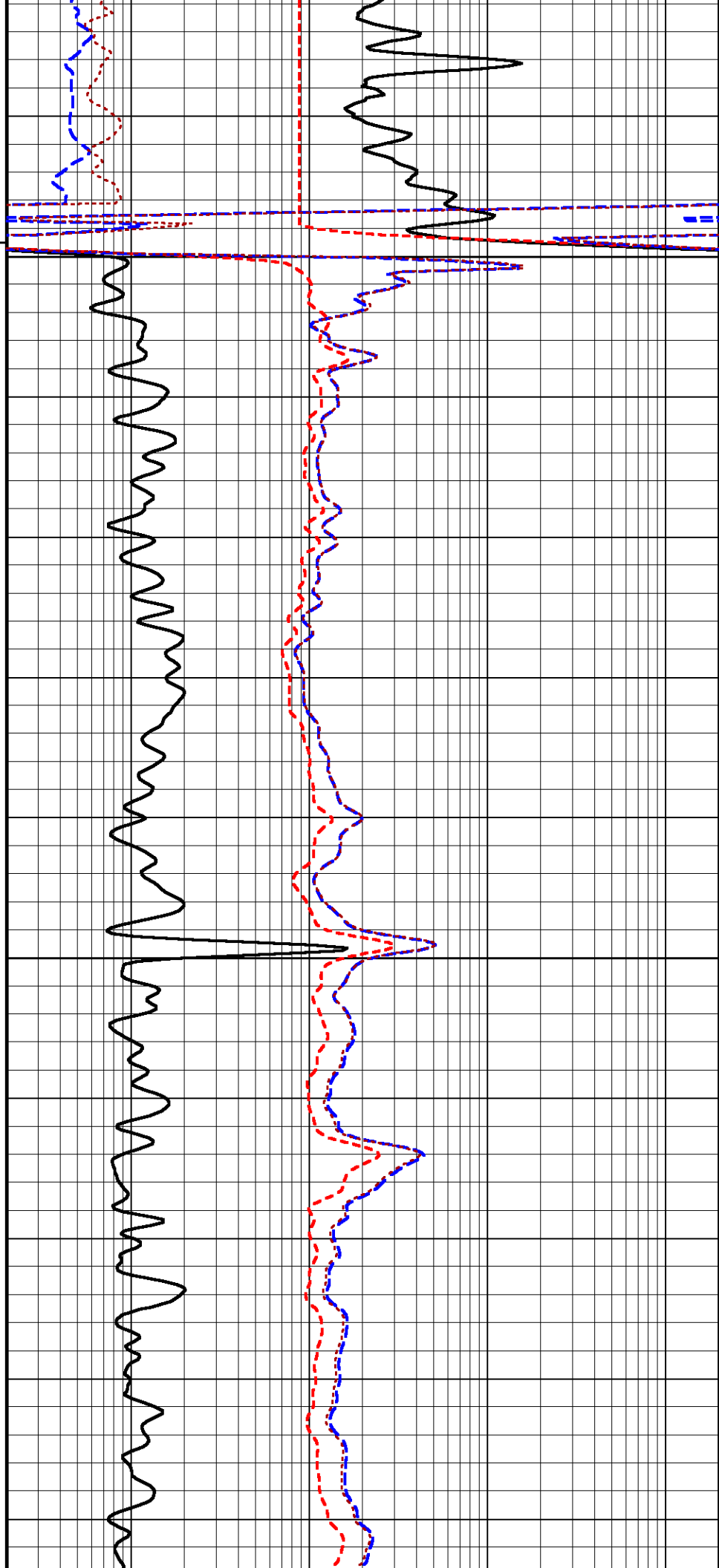
30°
Casing
Shoe
250

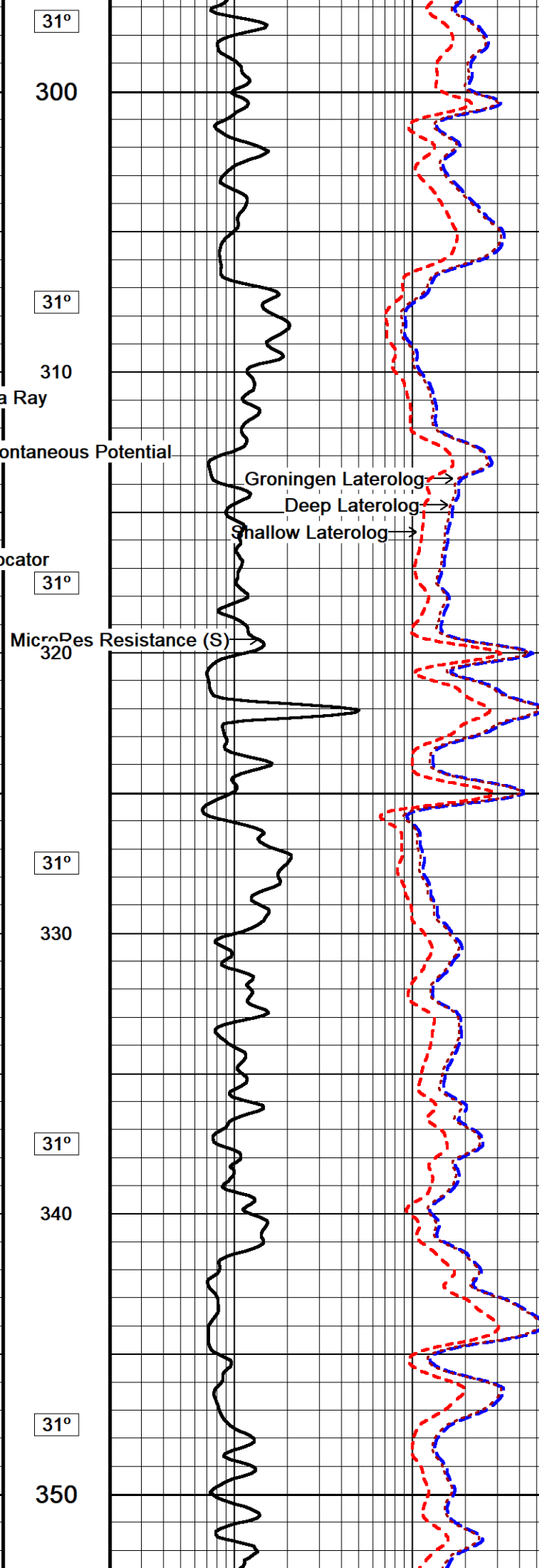
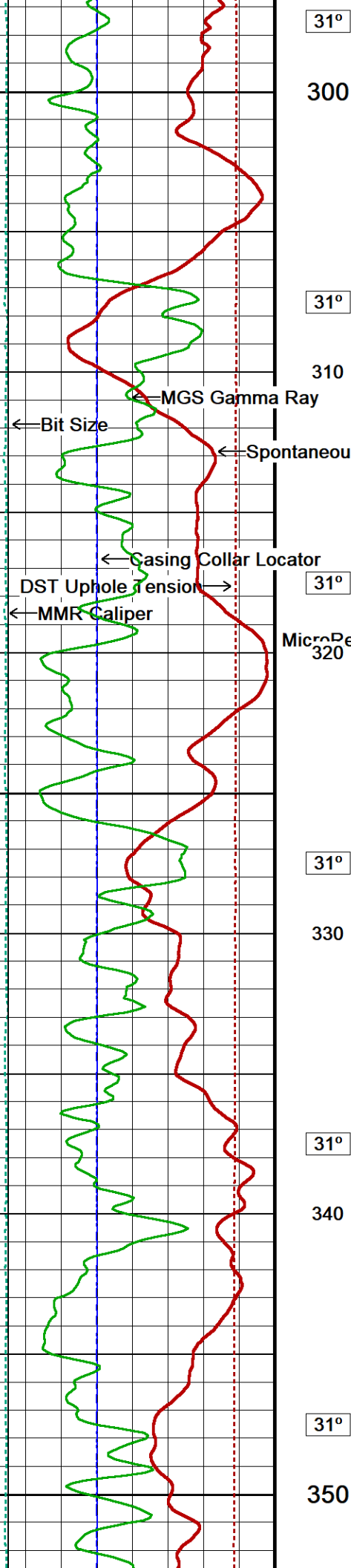
30°
260

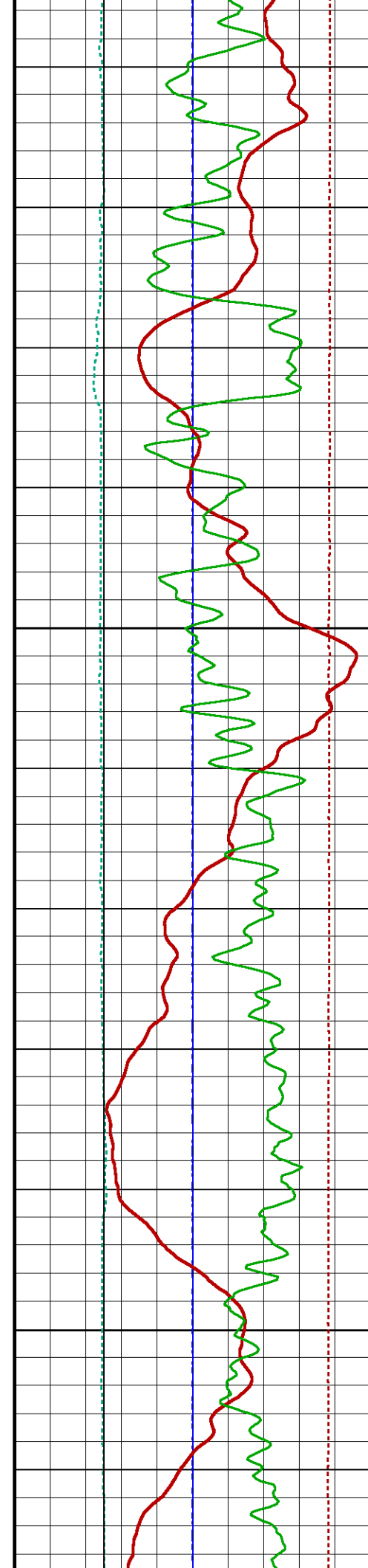
30°
270

30°
280

31°
290







31°

360

31°

370

31°

380

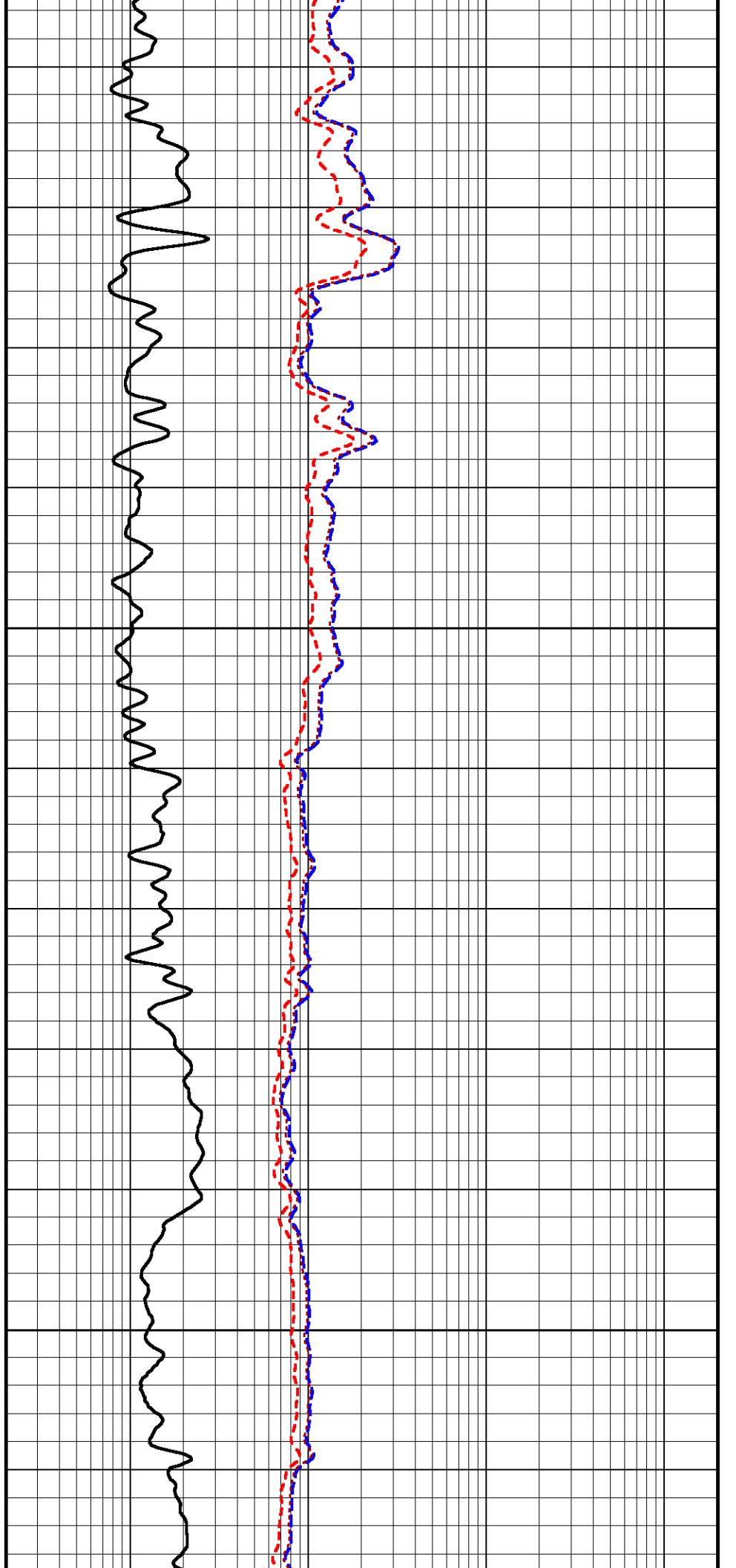
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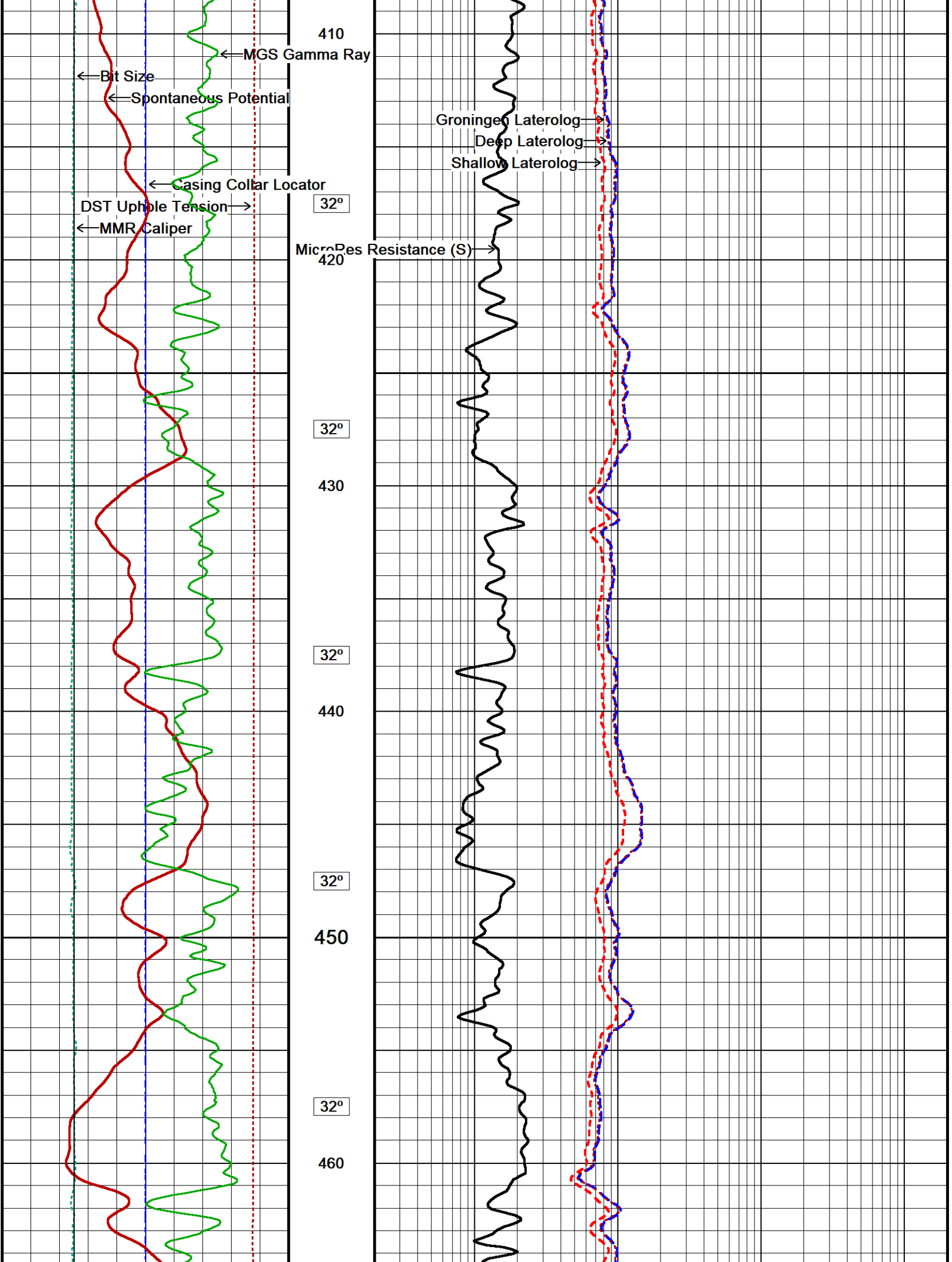
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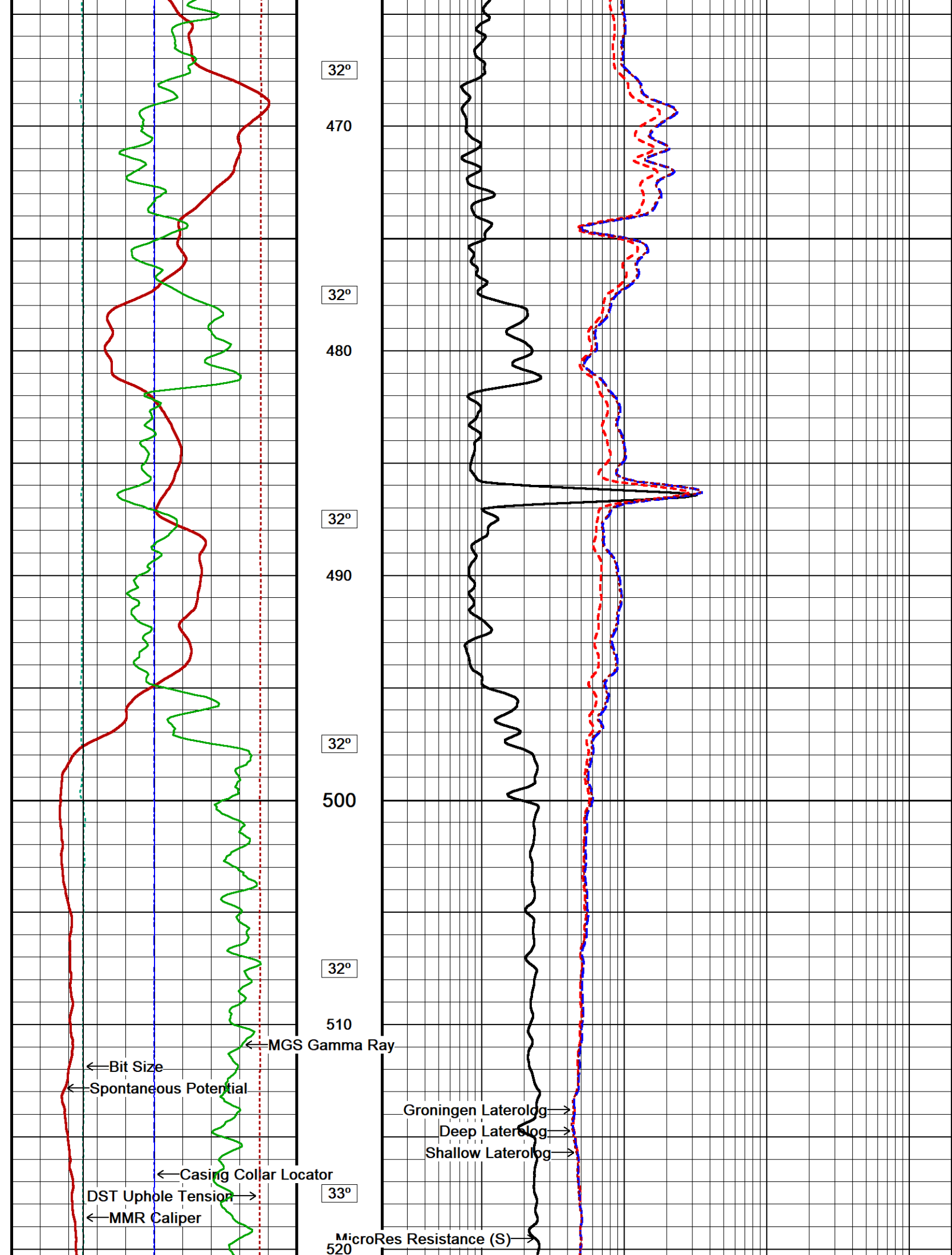
31°

400

31°









33°

530

33°

540

33°

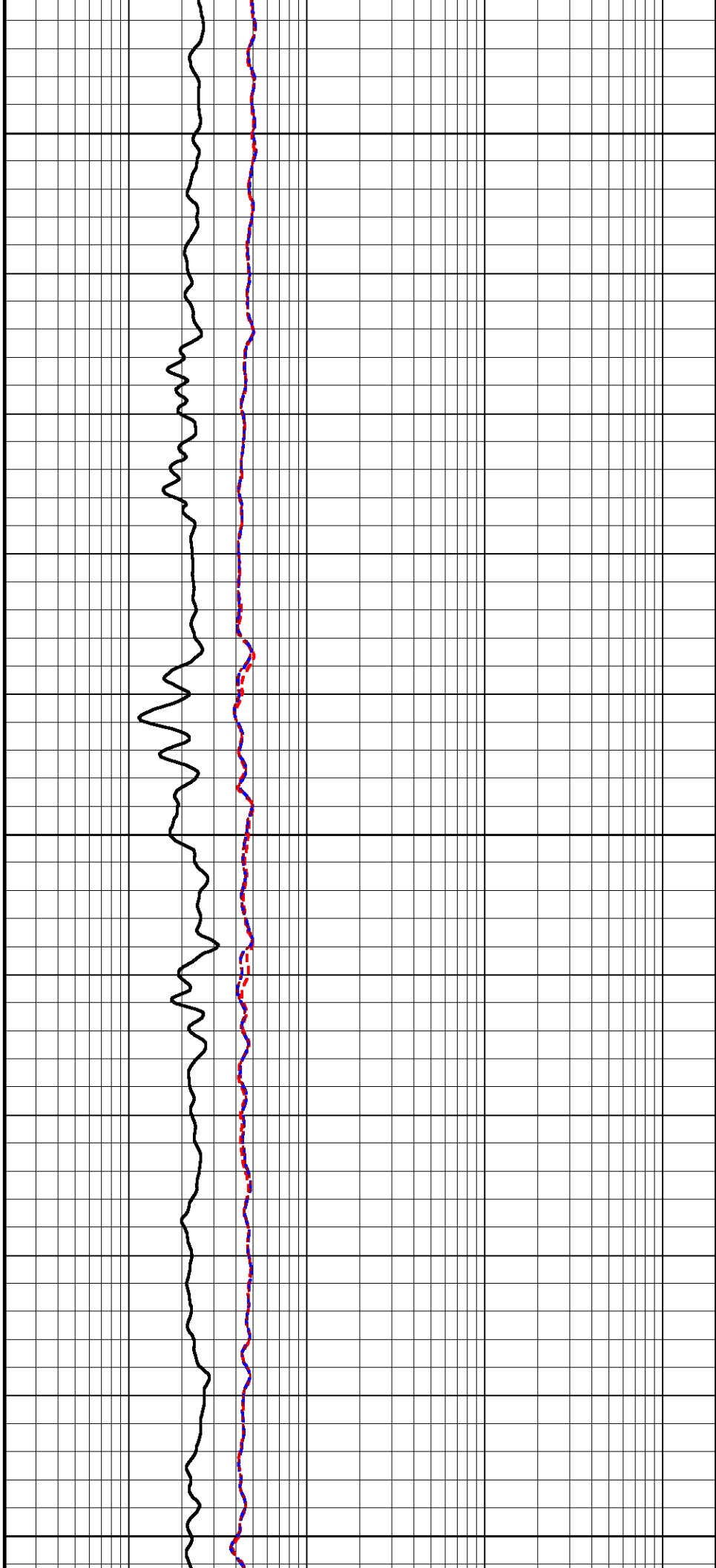
550

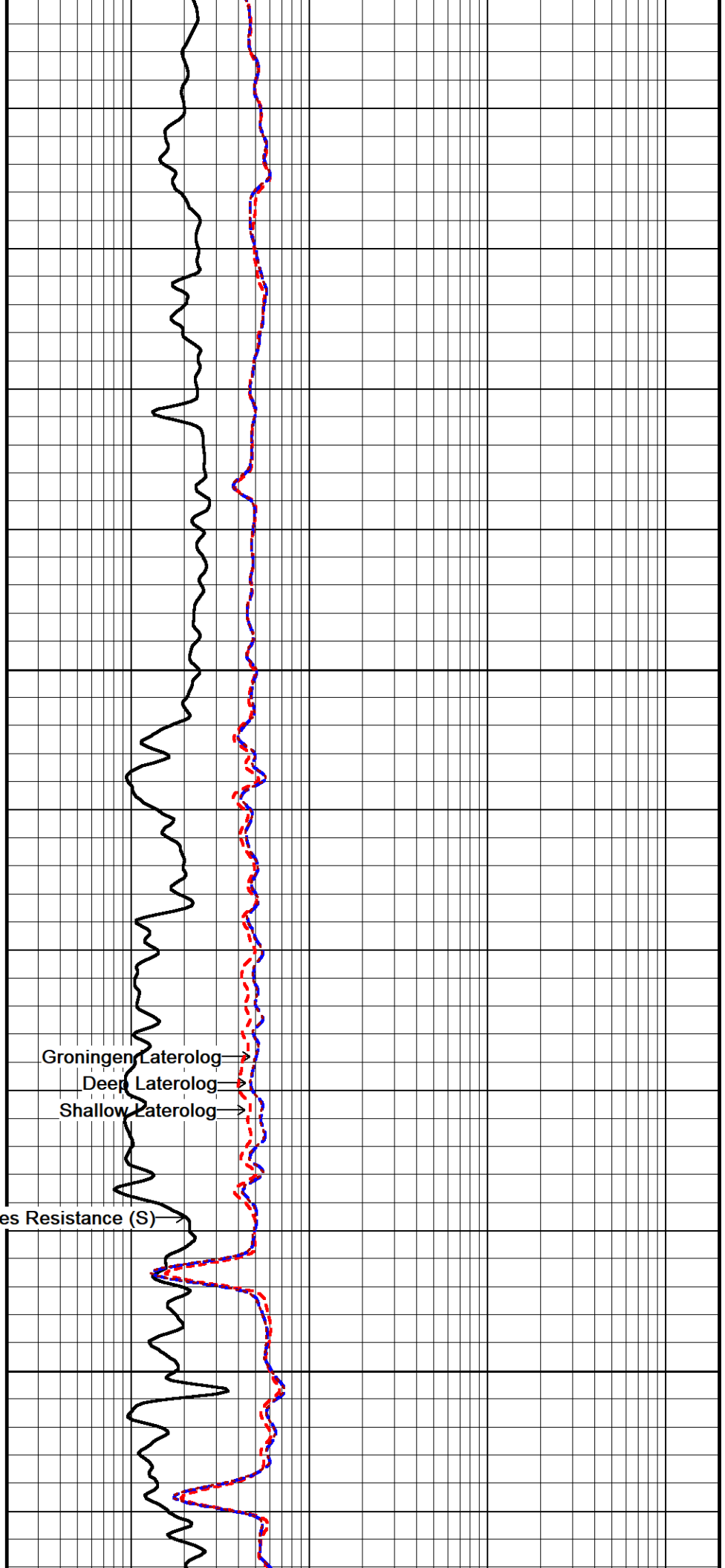
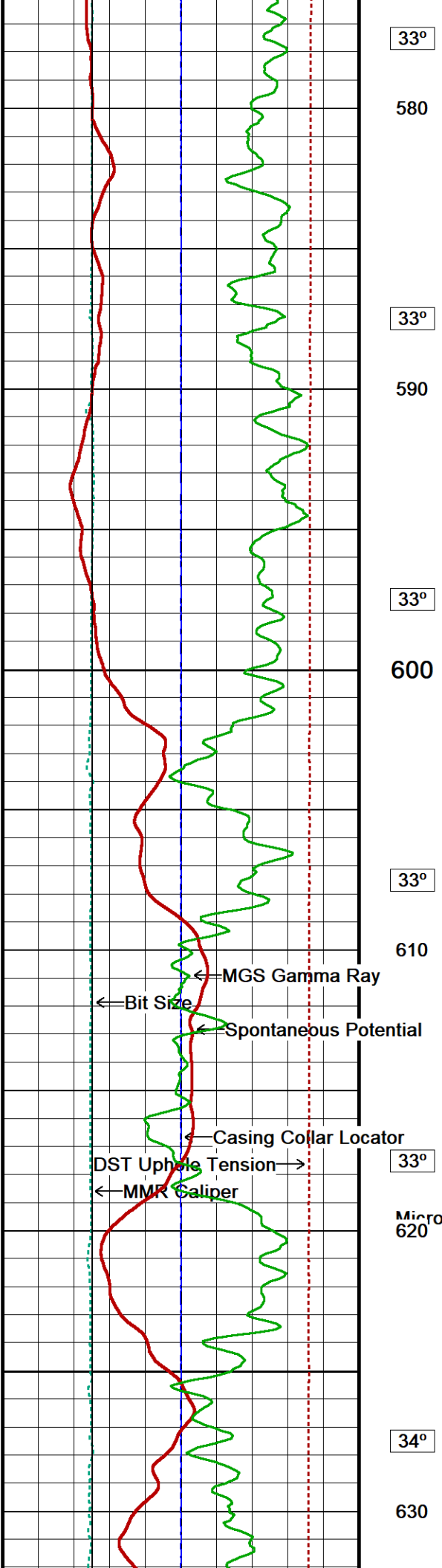
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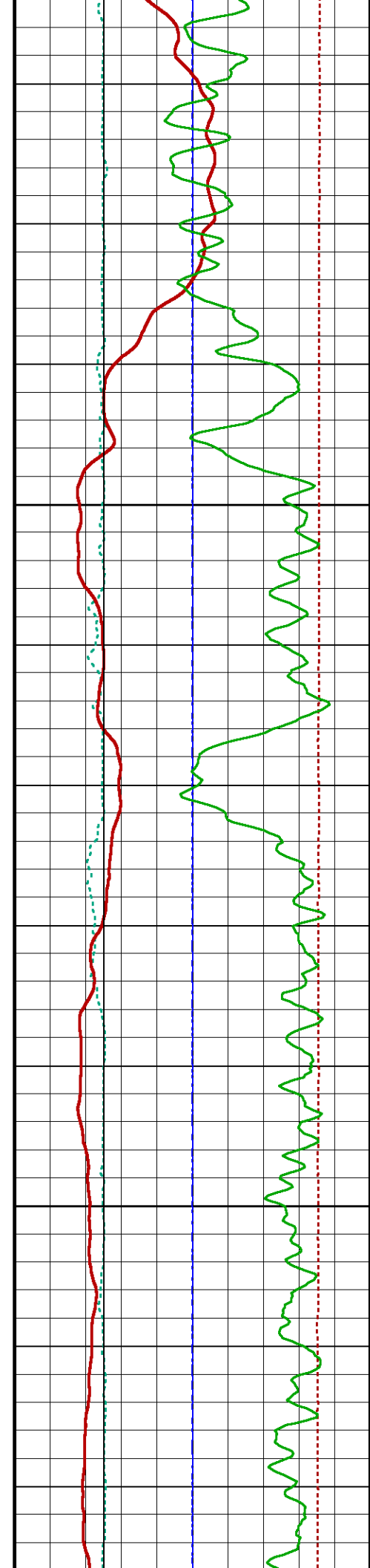
560

33°

570







34°

640

34°

650

34°

660

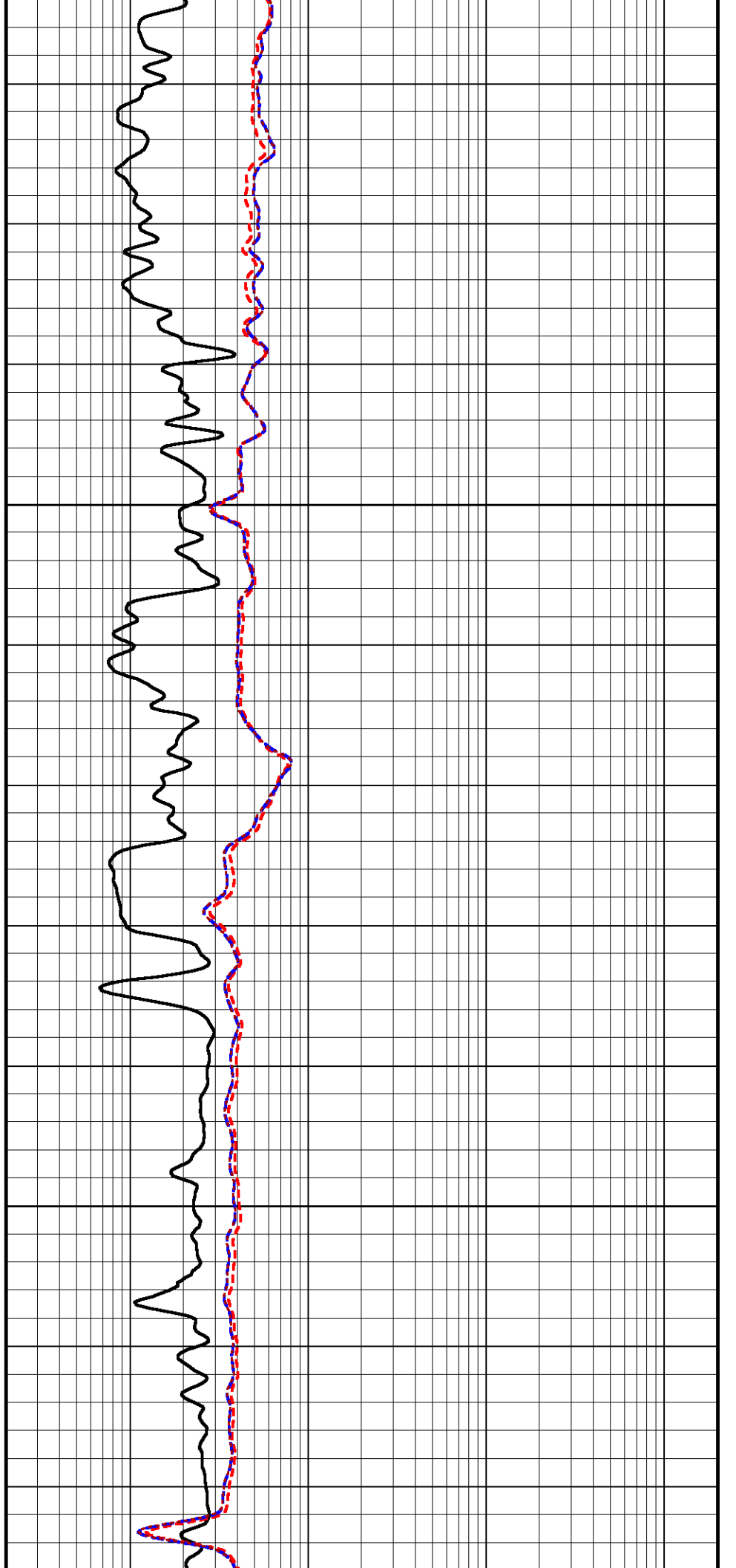
34°

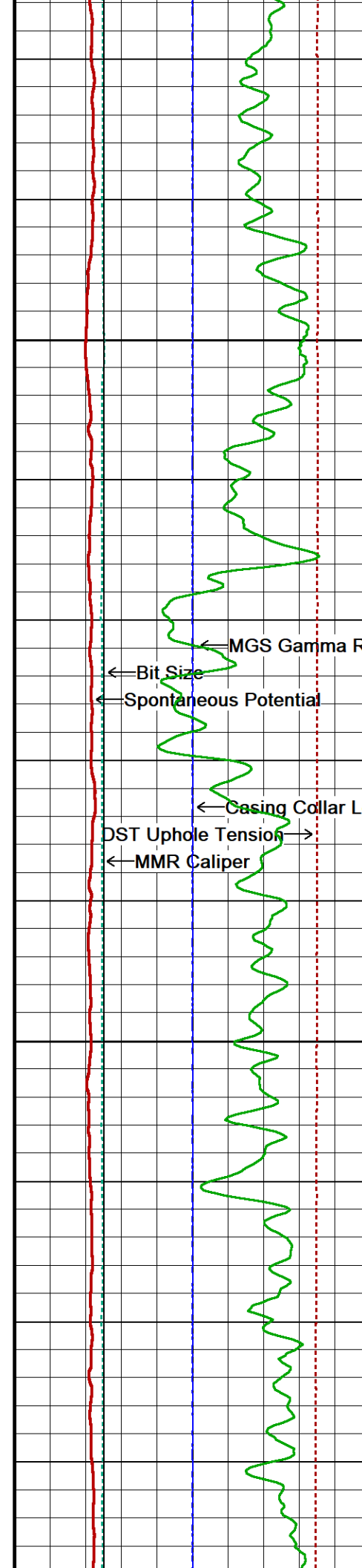
670

34°

680

34°

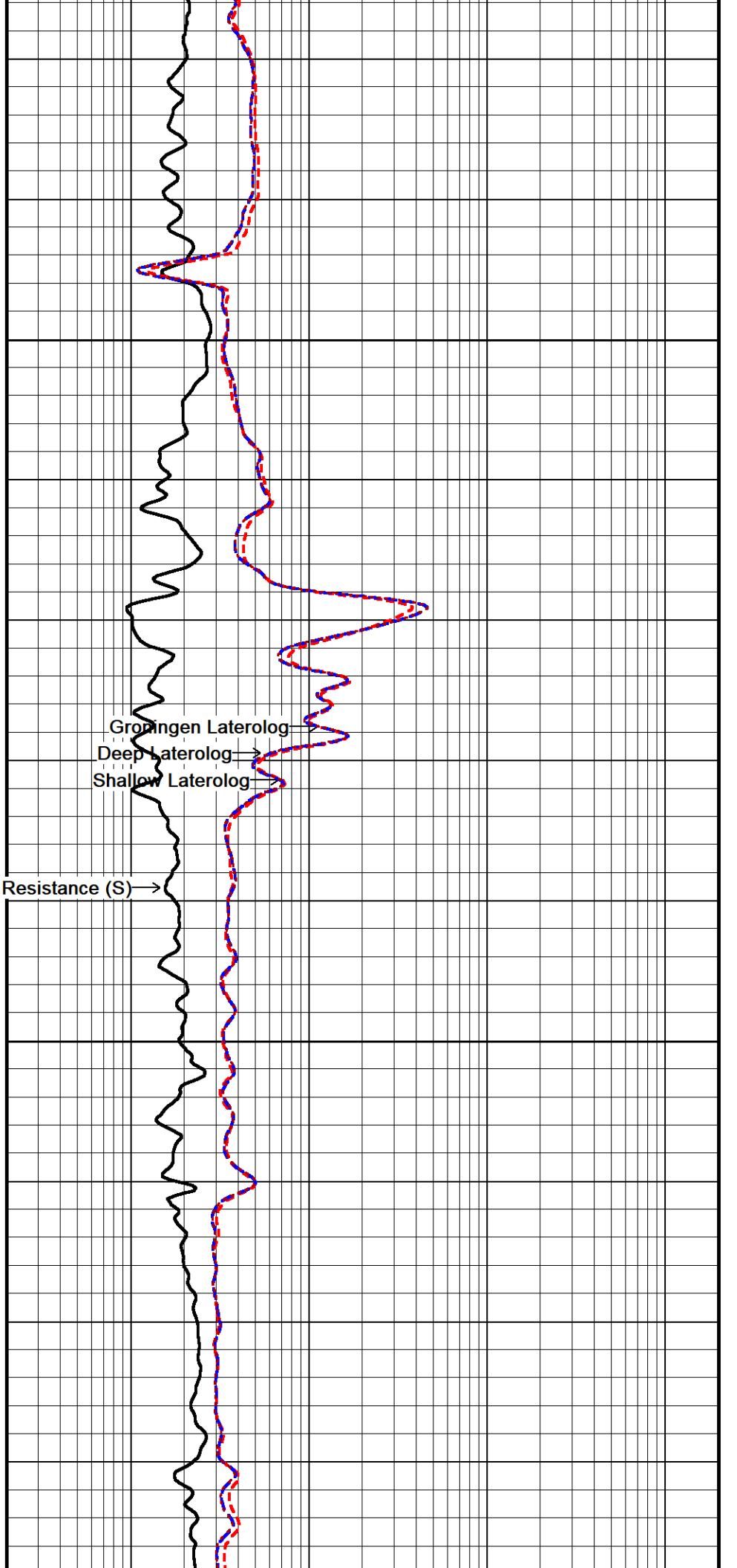




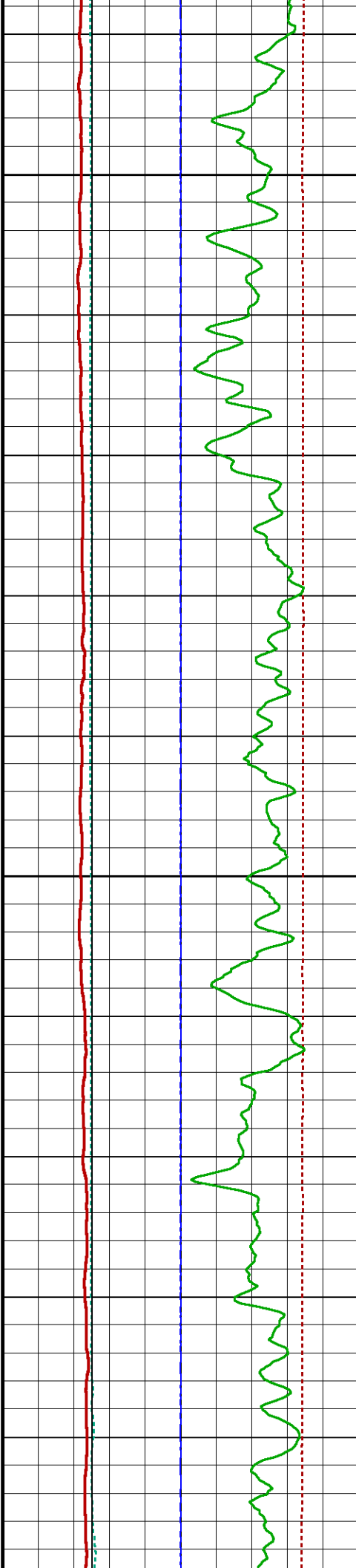
690
34°
700
35°
710
35°
720
35°
730
35°
740

← Bit Size
← Spontaneous Potential
← MGS Gamma Ray
← Casing Collar Locator
DST Uphole Tension →
← MMR Caliper

MicroRes Resistance (S) →



Groeninge Laterolog
Deep Laterolog
Shallow Laterolog



35°

750

35°

760

35°

770

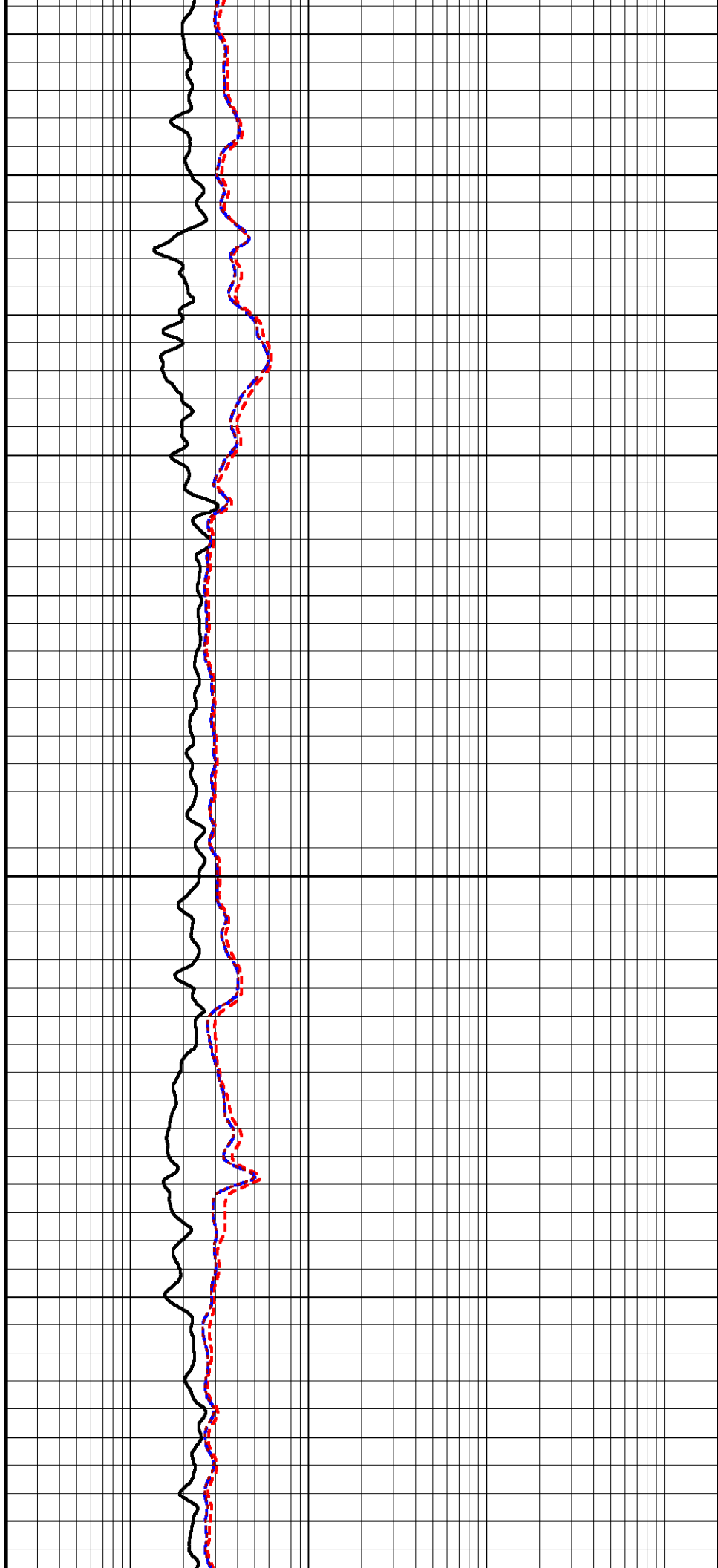
35°

780

35°

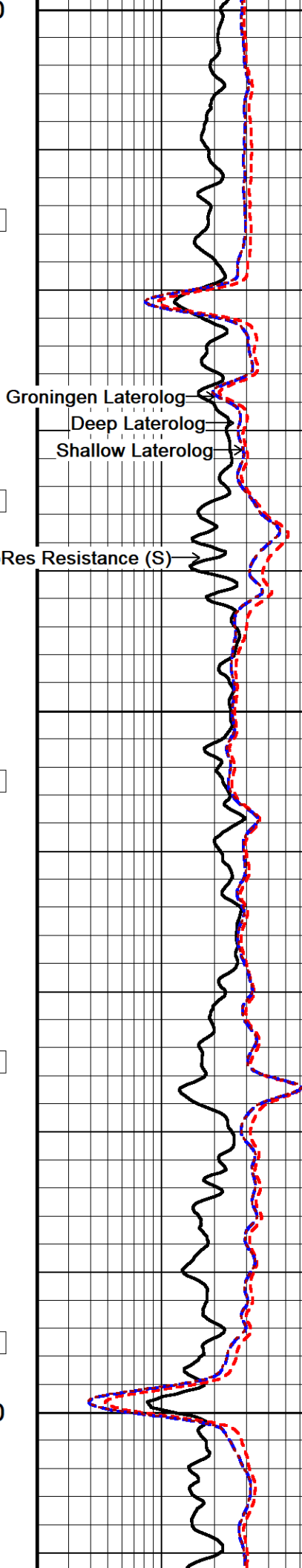
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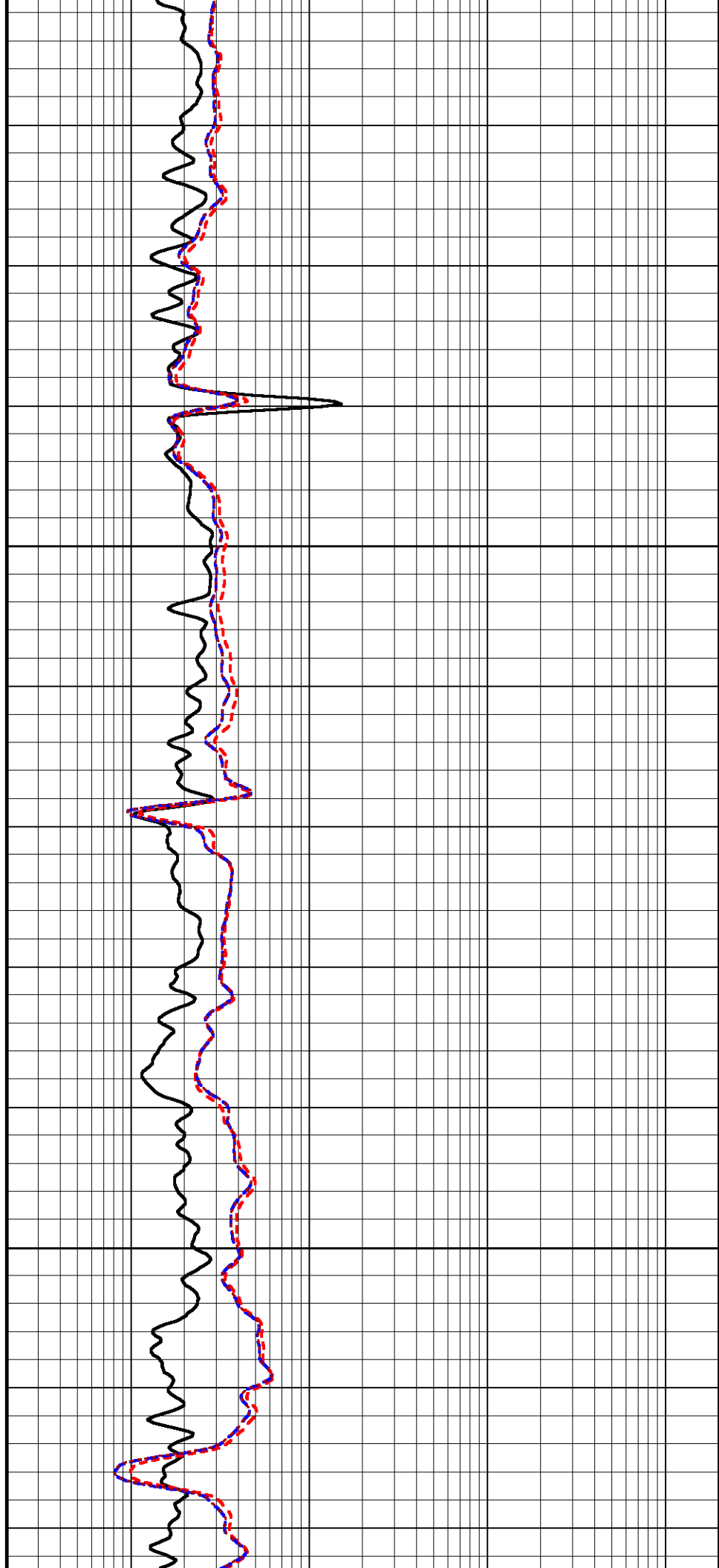
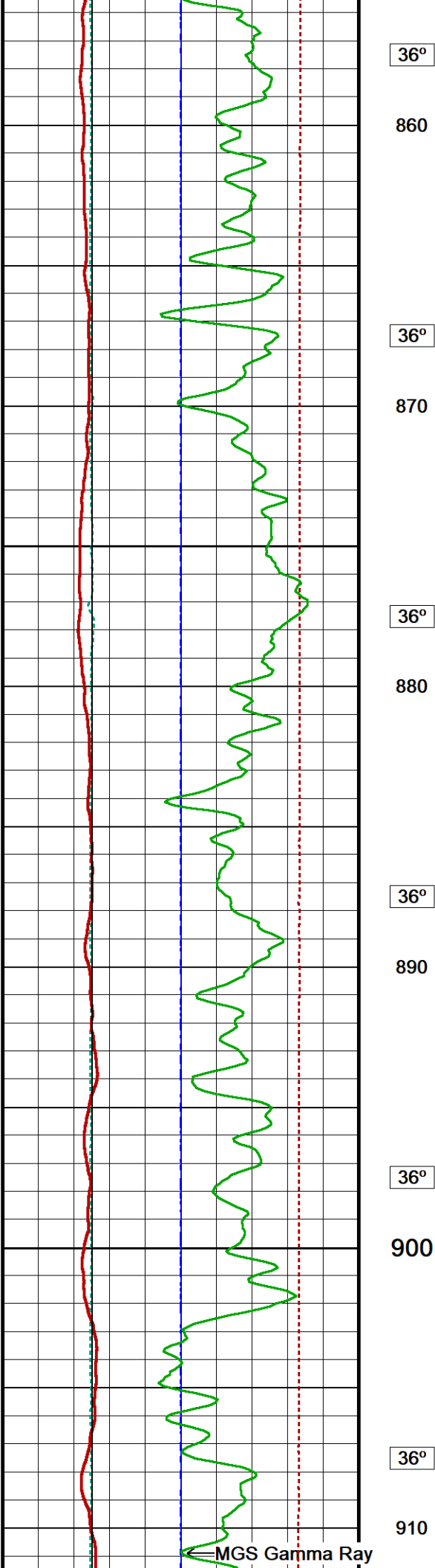
36°

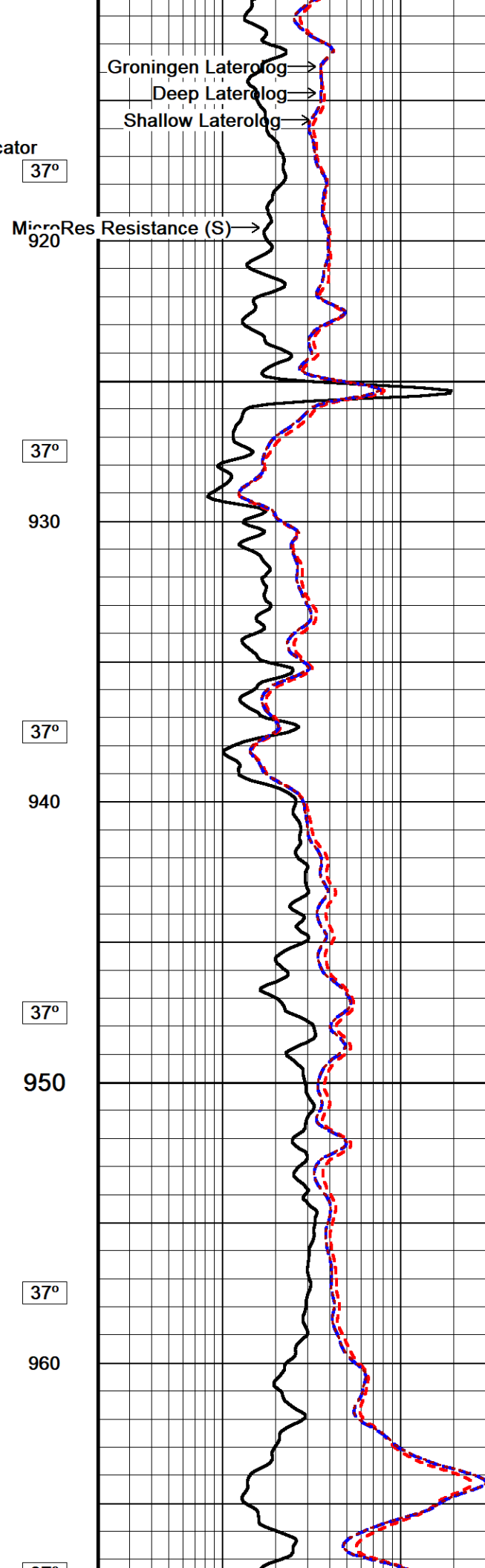
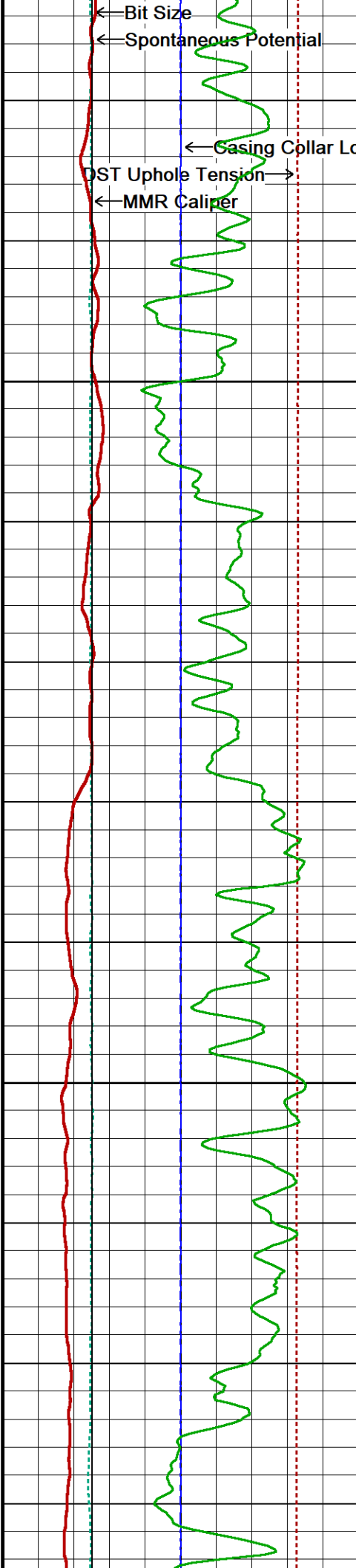


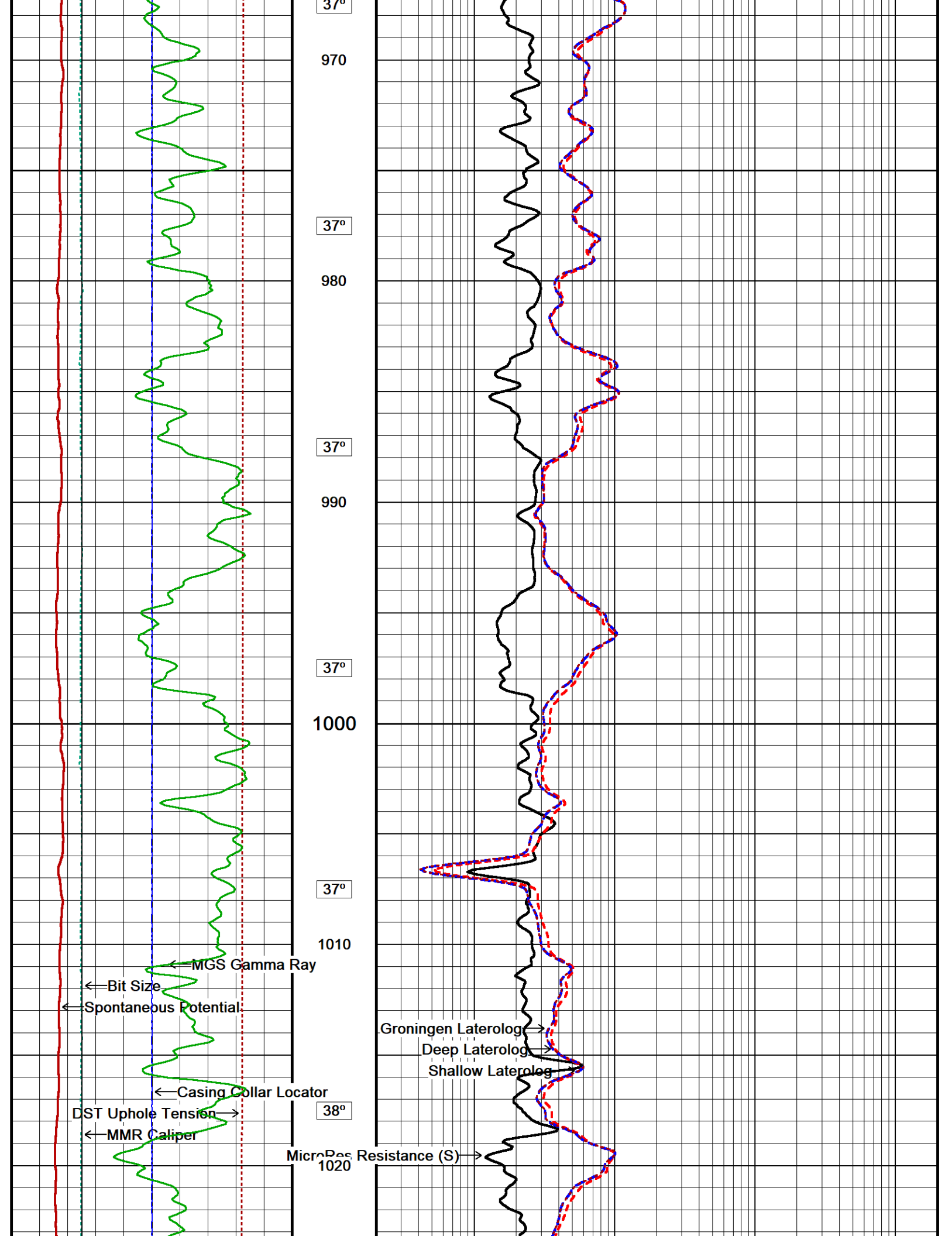


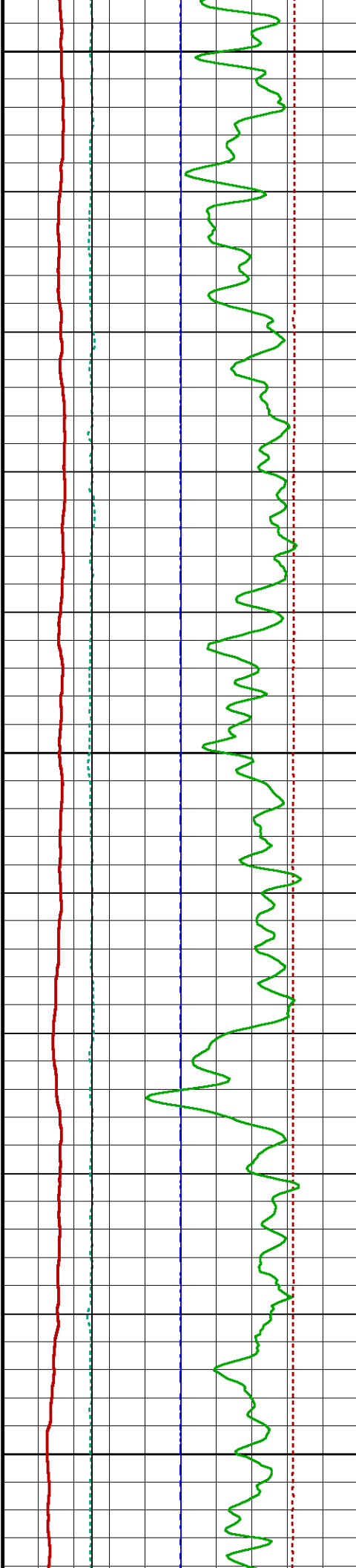
800
36°
810
36°
820
36°
830
36°
840
36°
850



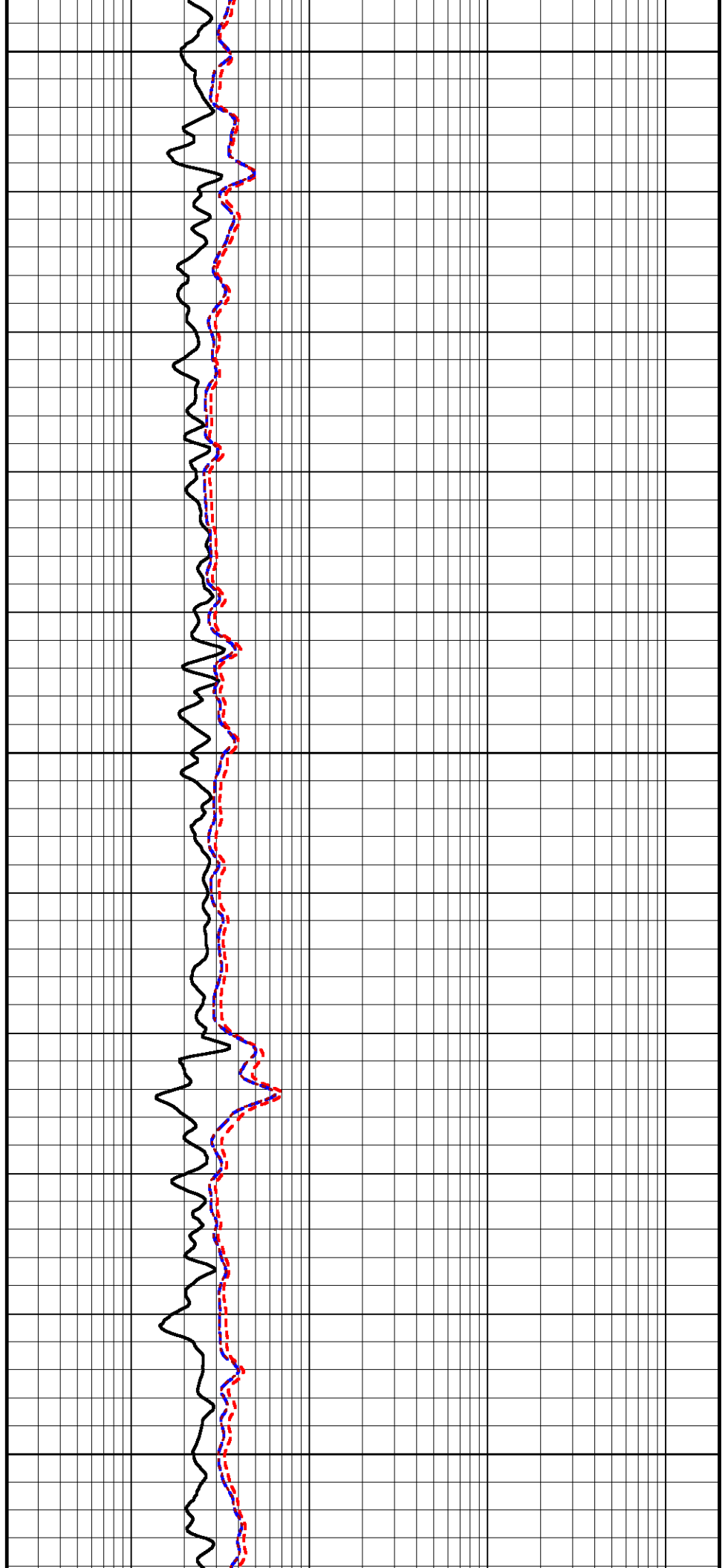


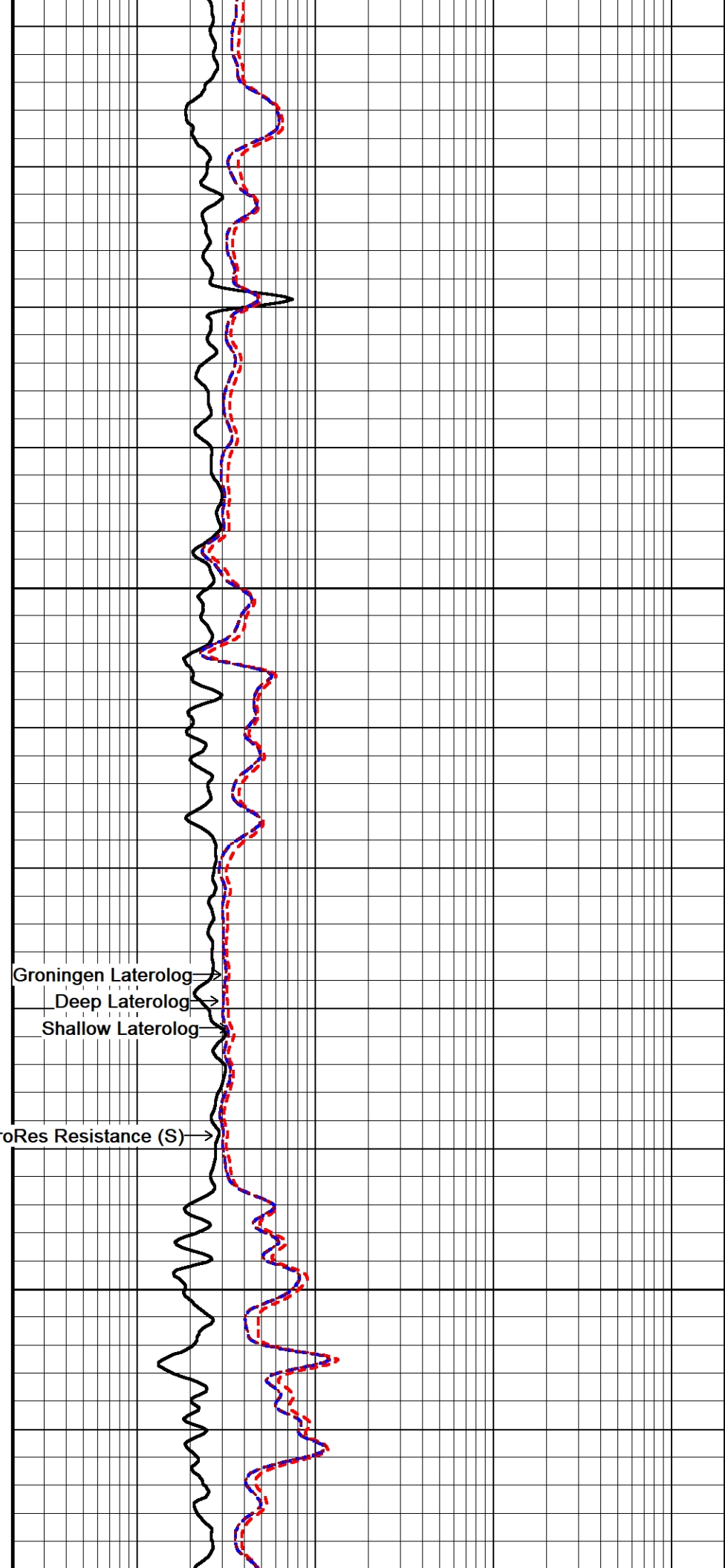
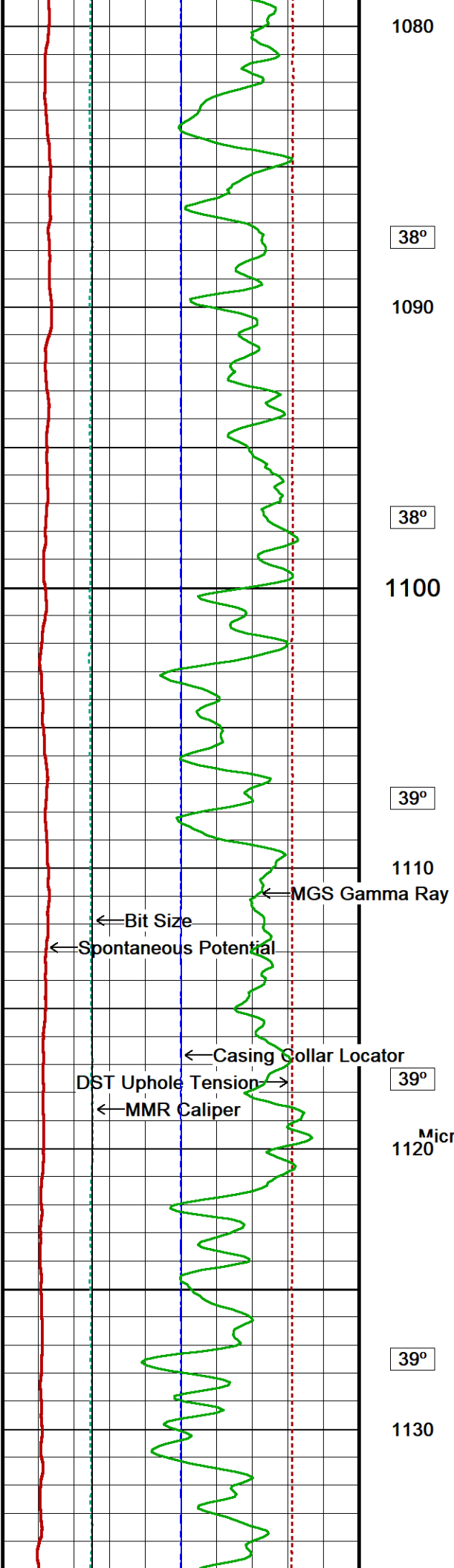


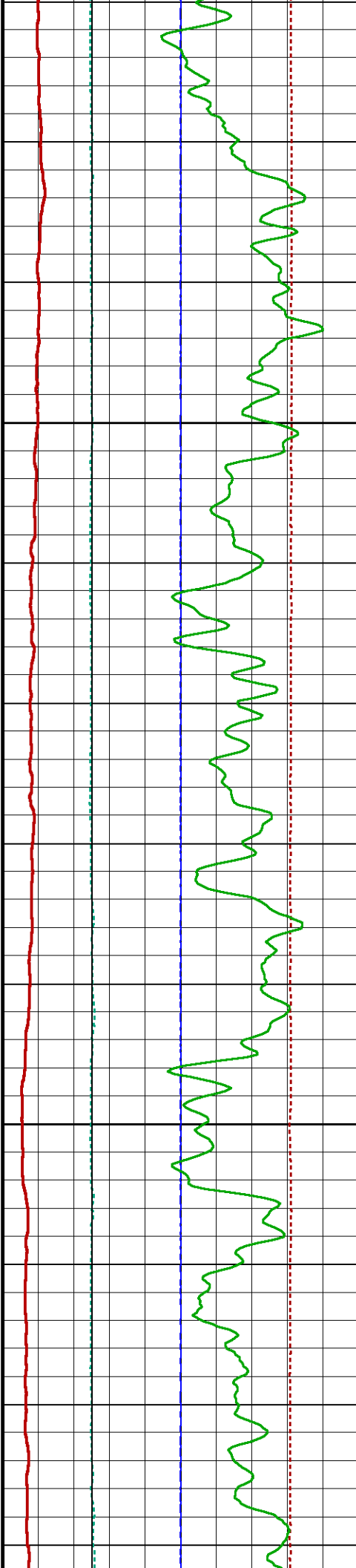




38°
1030
38°
1040
38°
1050
38°
1060
38°
1070
38°







39°

1140

39°

1150

39°

1160

39°

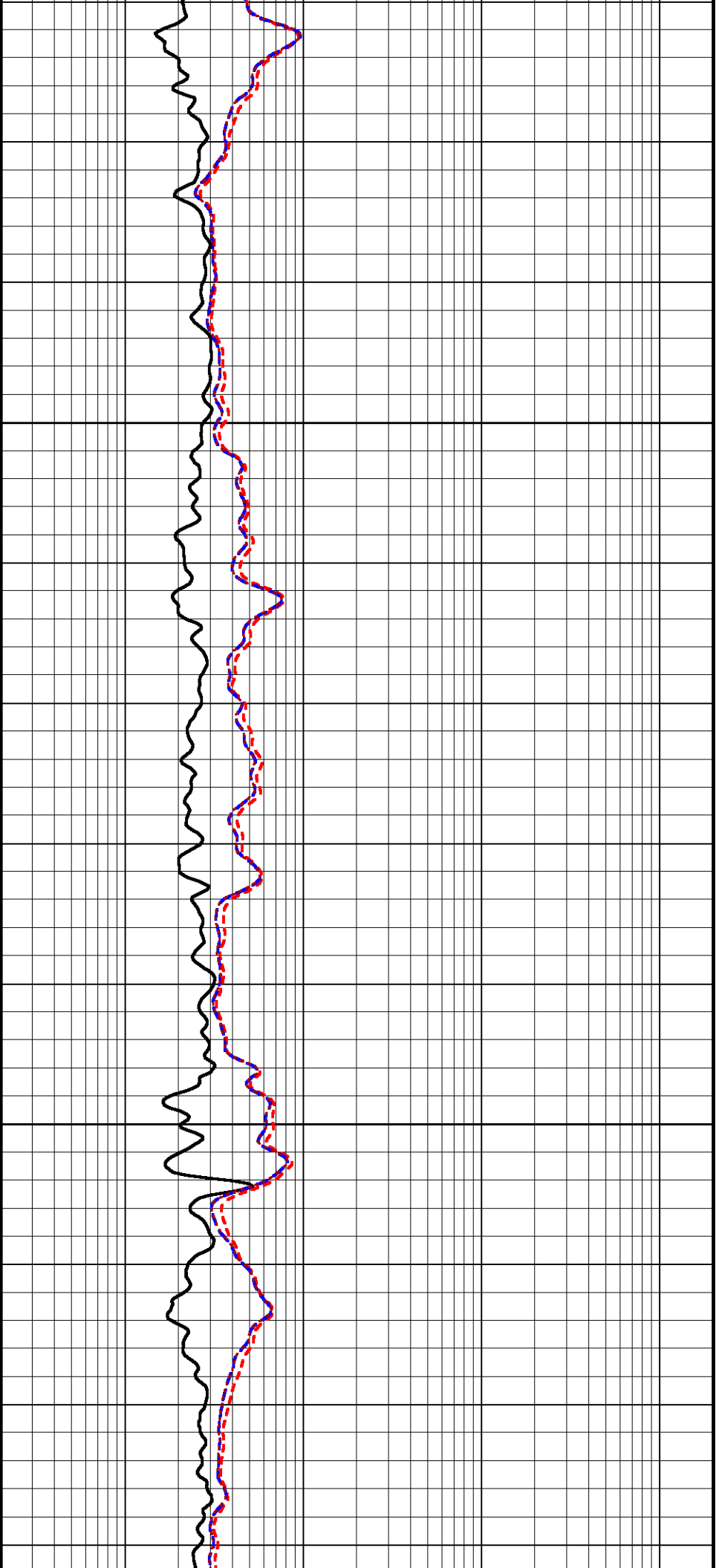
1170

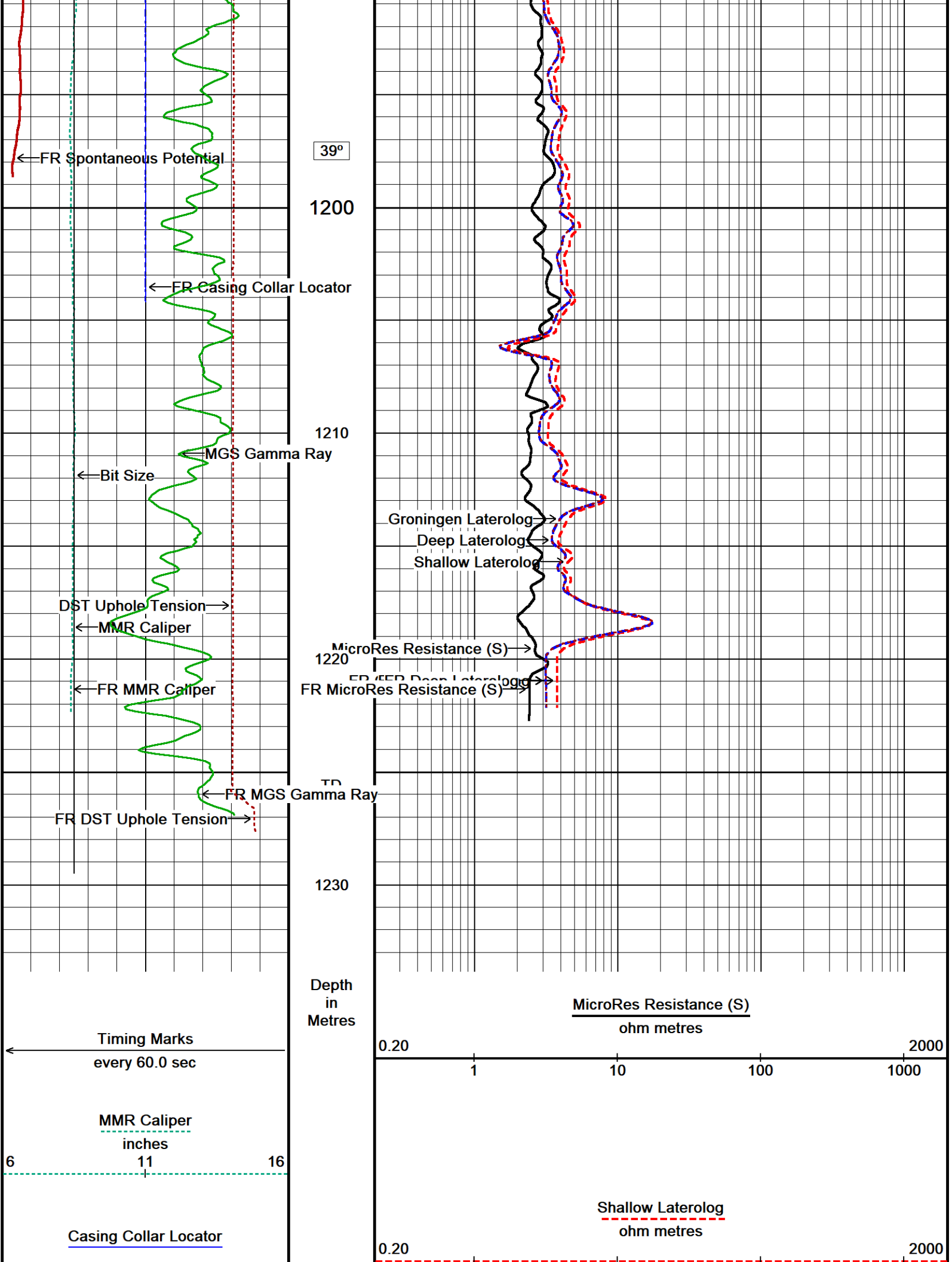
39°

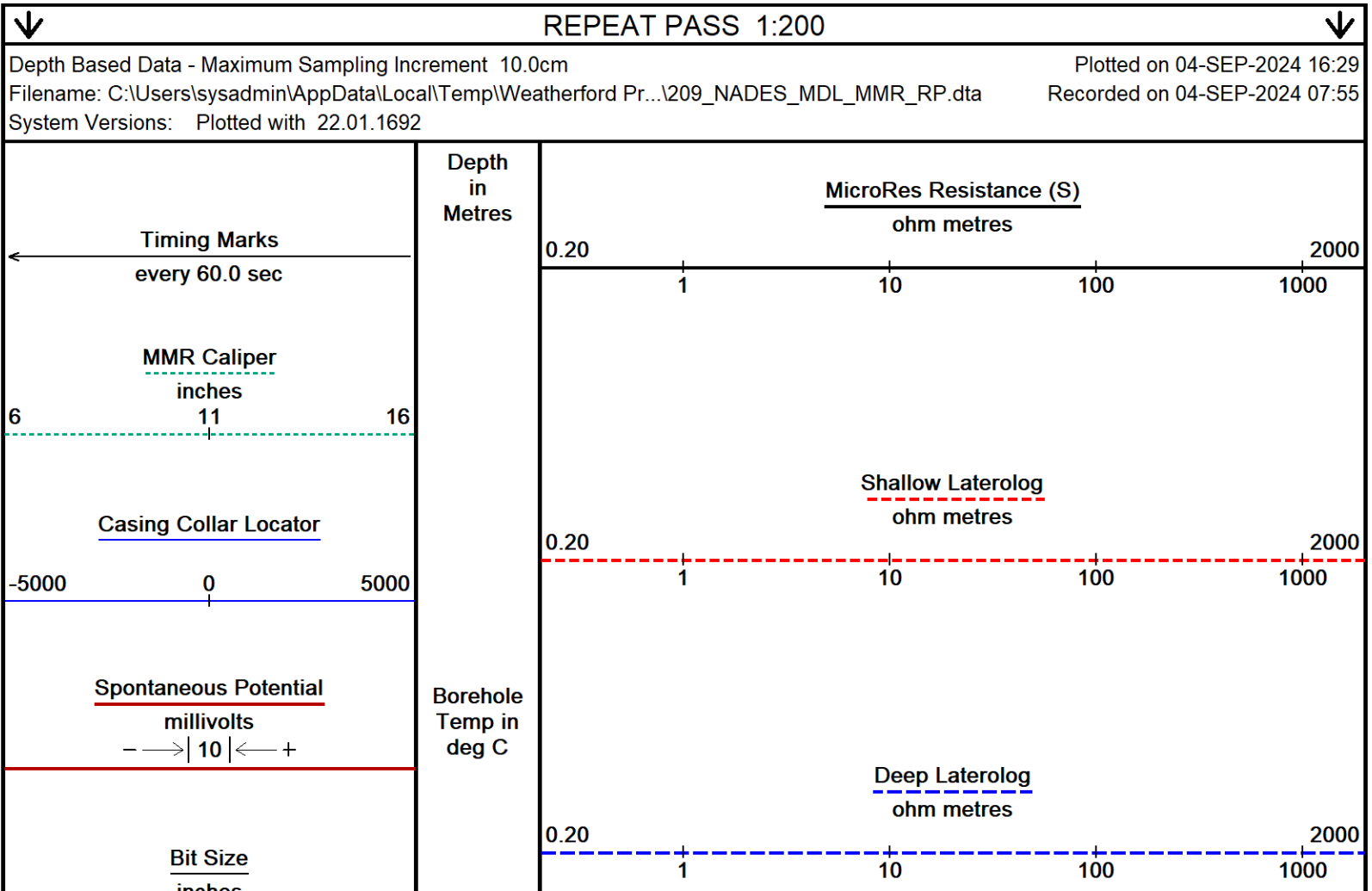
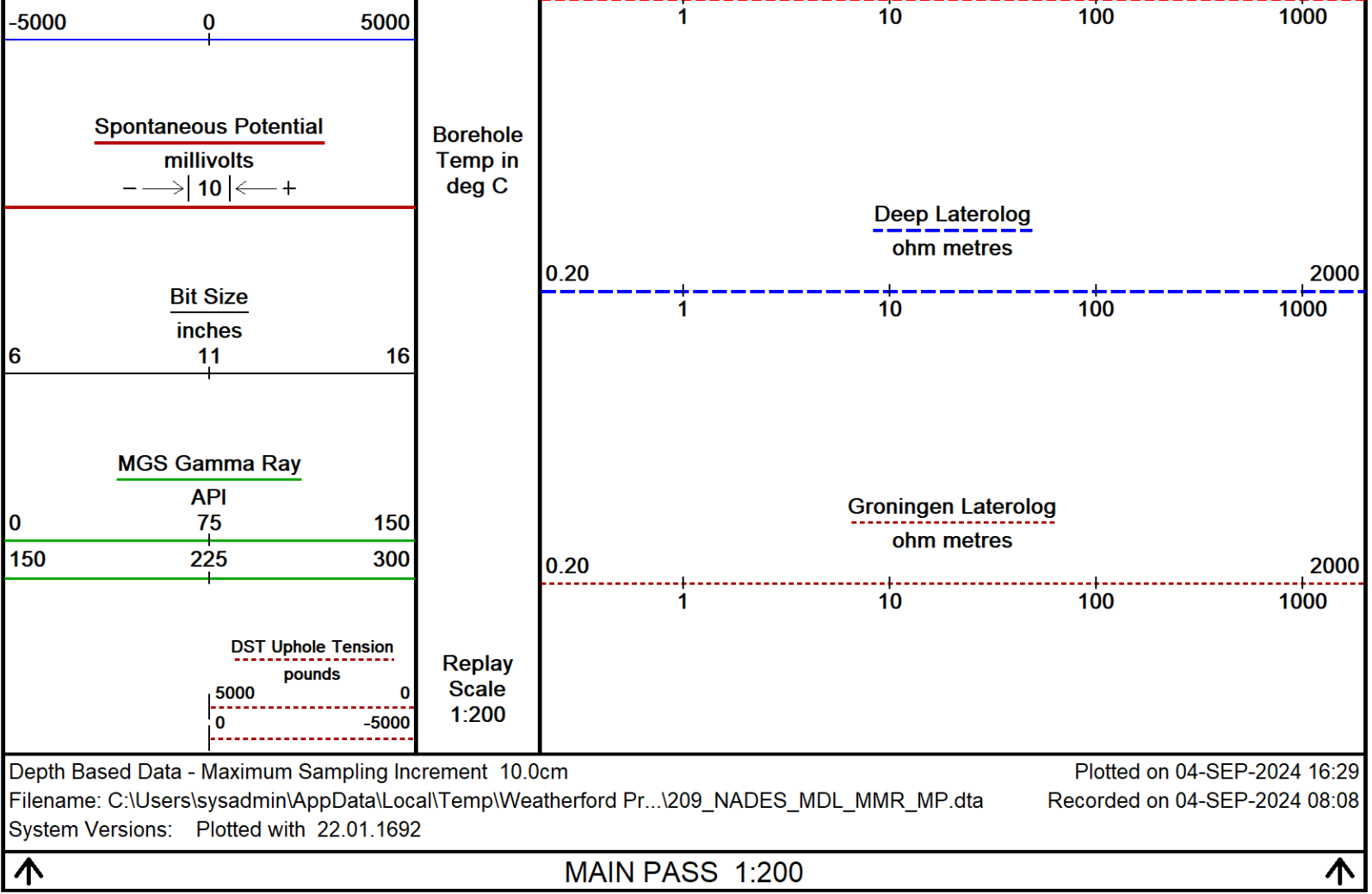
1180

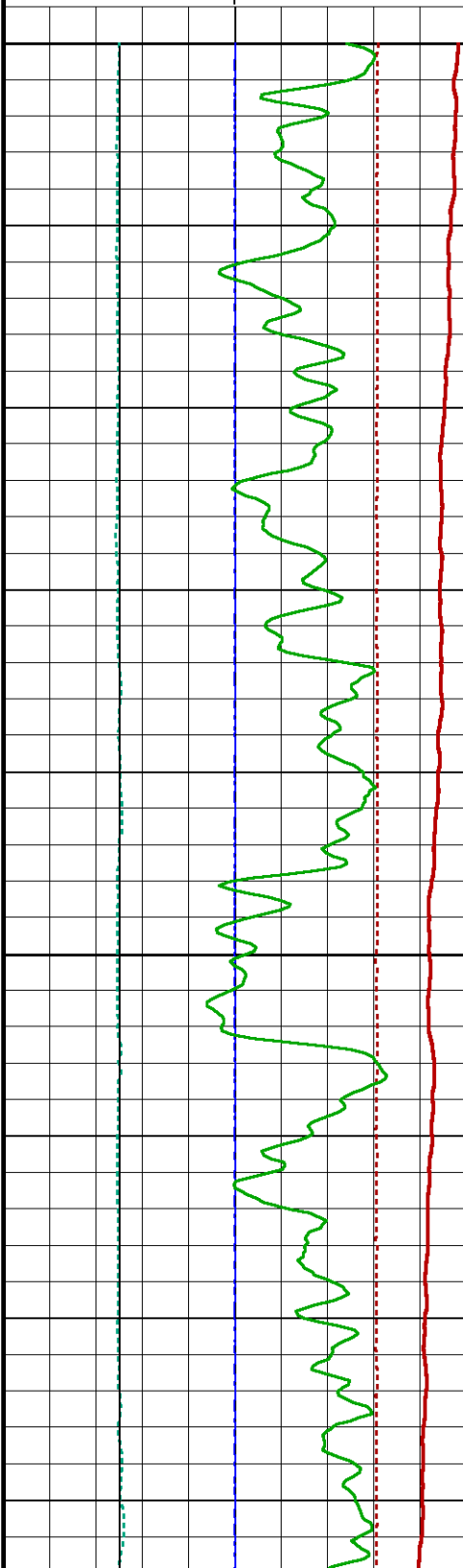
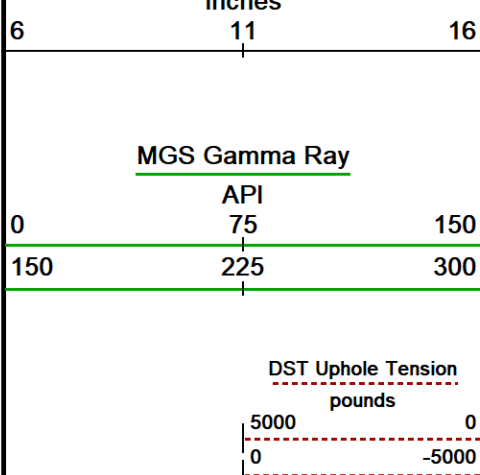
39°

1190









Replay
Scale
1:200

1150

39°

1160

39°

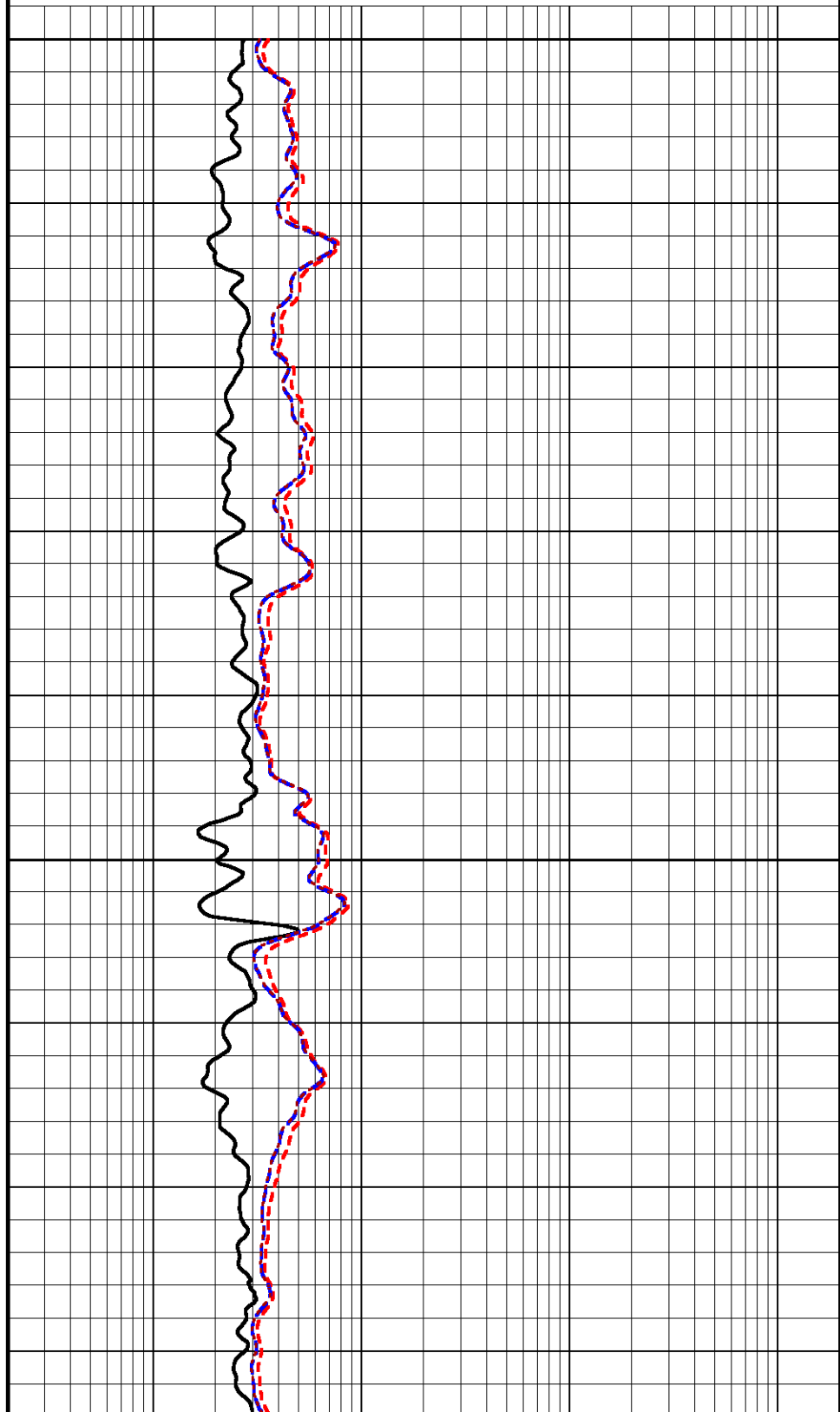
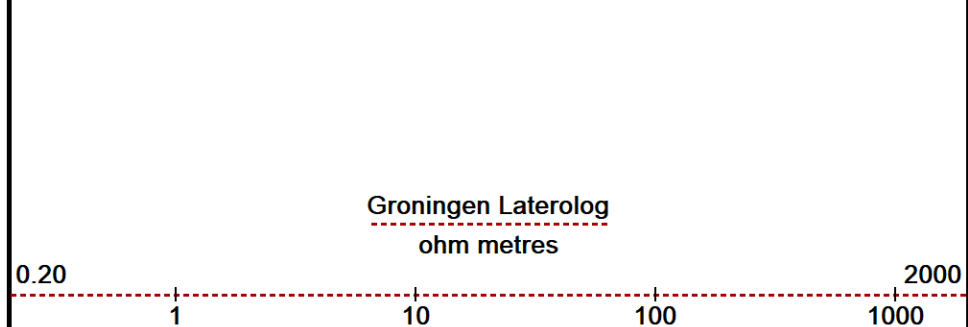
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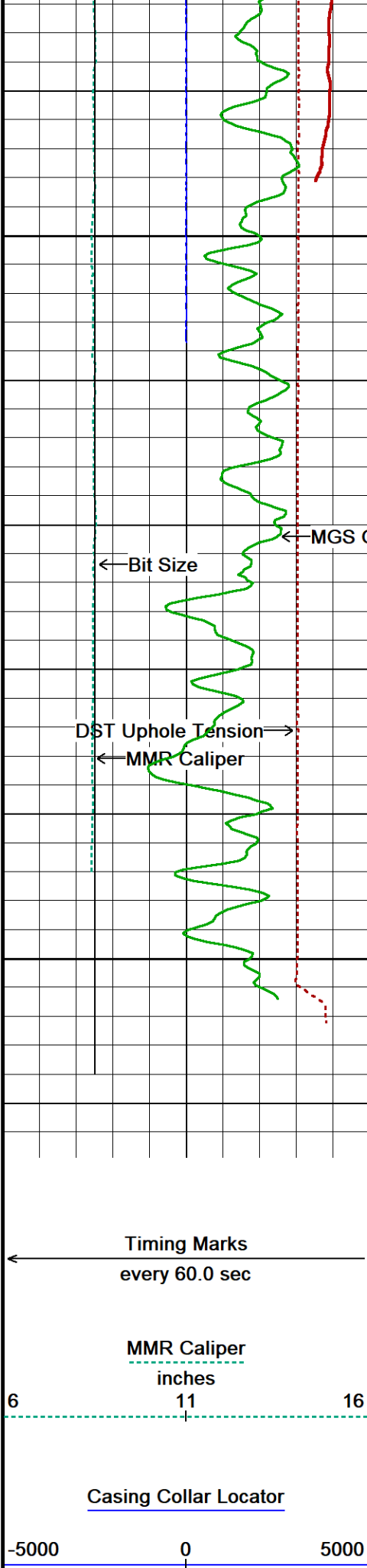
39°

1180

38°

1190





38°

1200

1210
MGS Gamma Ray

← Bit Size

DST Uphole Tension →

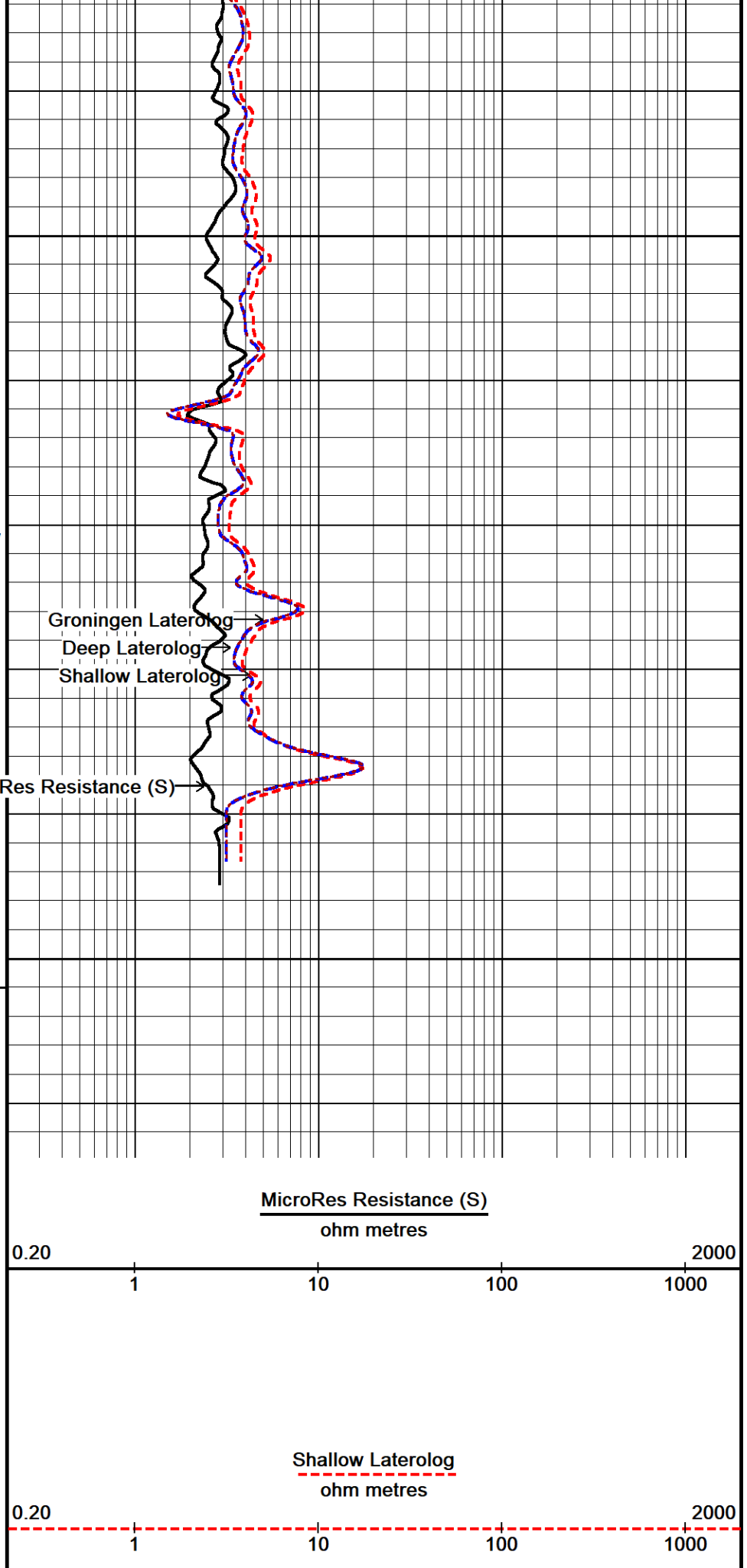
← MMR Caliper

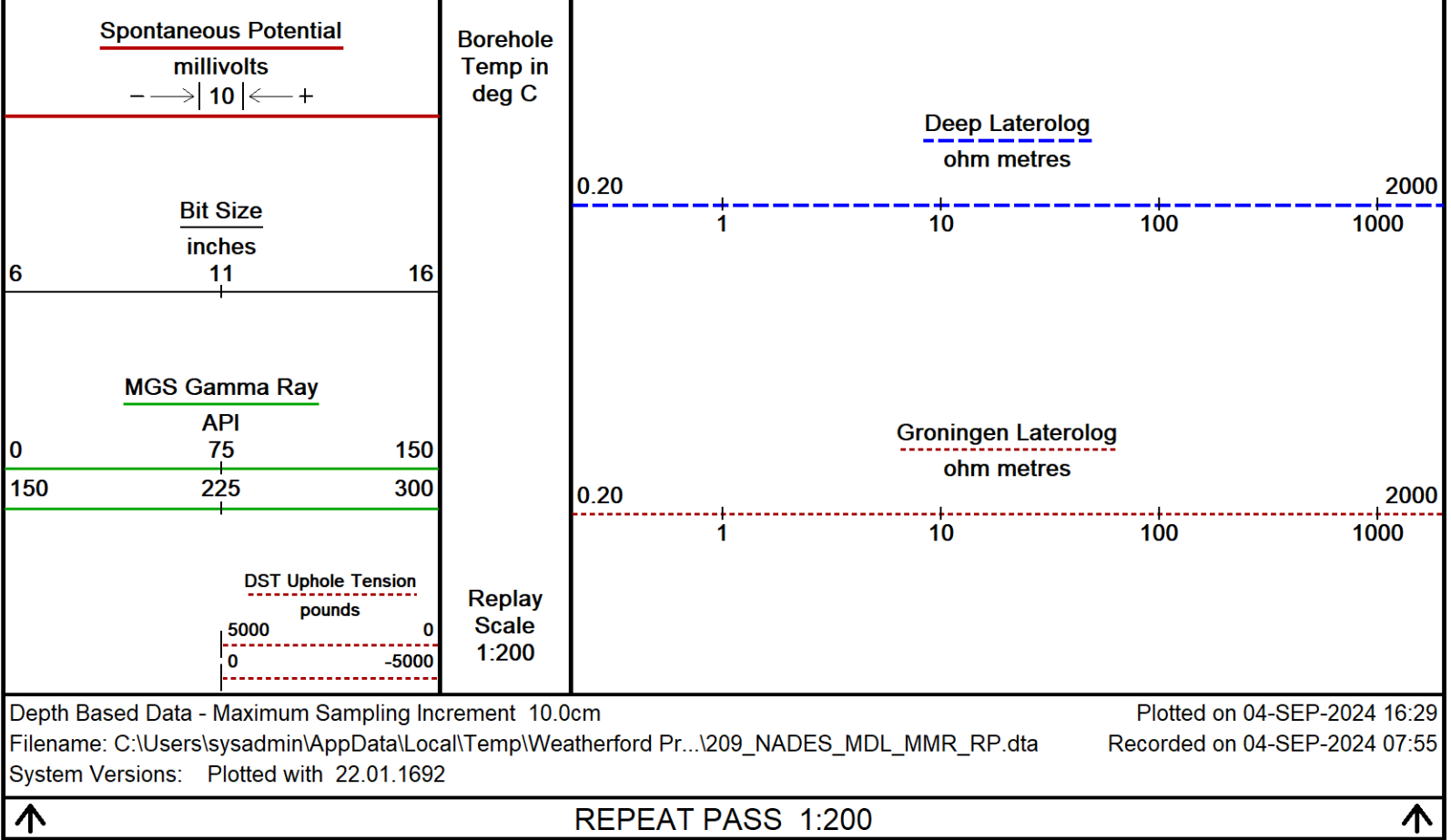
MicroRes Resistance (S)
1220

TD

1230

Depth
in
Metres





BEFORE SURVEY CALIBRATION			
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView90\0\209_NADES_RES_CALB.dta			
General Constants All 000		Last Edited on 04-SEP-2024,06:18	
General Parameters			
Mud Resistivity	0.150	ohm-metres	
Mud Resistivity Temperature	25.000	degrees C	
Water Level	0.000	metres	
Borehole Fluid Processing	Water Level Switch		
Hole/Annular Volume and Differential Caliper Parameters			
HVOL Method	XY Caliper		
HVOL Caliper 1	X Two Arm Caliper		
HVOL Caliper 2	Y Two Arm Caliper		
Annular Volume Diameter	5.500	inches	
Caliper for Differential Caliper	None		
Rwa Parameters			
Porosity used	Wyllie Lmst Sonic Por.		
Resistivity used	Deep Laterolog		
RWA Constant A	0.620		
RWA Constant M	2.150		
SW/APOR Tool Source	0.000		
High Resolution Temperature Calibration MCG-E.A 531		Field Calibration on 03-SEP-2024,09:41	
	Measured	Calibrated(Deg C)	
Lower	0.00	0.00	
Upper	100.00	100.00	
High Resolution Temperature Constants MCG-E.A 531		Last Edited on 03-SEP-2024,09:41	
Pre-filter Length		11	
Accelerometer Parameters MBN-C.A 55			
Date Of Last Accelerometer Calibration		20-NOV-2023,13:27	
X Accelerometer		Y Accelerometer	
		Z Accelerometer	

Slope	X Accelerometer	-1.084180	Y Accelerometer	-1.103980	Z Accelerometer	-1.114574
Offset		0.002710		0.006361		0.008729
Accelerometer Constants MBN-C.A 55						
Accelerometer Calibrator Number		000				
Accelerometer Temperature Characterisation						
X Accelerometer						
Serial Number		0				
Calibration Date		01-Jan-1998				
		B0	B1	B2	B3	
Bias(g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
		SF0	SF1	SF2	SF3	
Scale Factor(mA/g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
Y Accelerometer						
Serial Number		0				
Calibration Date		01-Jan-1998				
		B0	B1	B2	B3	
Bias(g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
		SF0	SF1	SF2	SF3	
Scale Factor(mA/g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
Z Accelerometer						
Serial Number		0				
Calibration Date		01-Jan-1998				
		B0	B1	B2	B3	
Bias(g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
		SF0	SF1	SF2	SF3	
Scale Factor(mA/g)		0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	

Navigation Constants MBN-C.A 55				Last Edited on 04-SEP-2024,06:19		
Magnetic Declination		6.53	degrees	East		

Magnetometer Parameters MBN-C.A 55						
Date Of Last Magnetometer Calibration		22-MAY-2024,11:04				
	X Magnetometer	Y Magnetometer		Z Magnetometer		
Slope	-1.000000	1.002324		1.004838		
Offset	0.006984	0.017764		-0.012985		

Magnetometer Constants MBN-C.A 55						
Magnetometer Calibrator Number		000				

SP Calibration MLE-D.A 283				Field Calibration on 13-AUG-2024 11:09		
	Measured		Calibrated (mV)			
Reference 1	98.7		100.0			
Reference 2	-98.0		-100.0			

Laterolog Calibration MLE-D.A 283				Base Calibration on 23-AUG-2024 13:47	
				Field Check on 04-SEP-2024 07:22	
		Resistor 1 (ohm)		Resistor 2 (ohm)	
		0.0		1000.0	
Base Calibration					
		Measured		Calibrated (ohm-m)	
Channel	Resistor 1	Resistor 2	Resistor 1	Resistor 2	
Shallow	0.0	1005.2	0.0	1284.4	
Deep	0.0	1005.1	0.0	795.7	
Groningen	0.0	1005.8	0.0	808.4	
Channel	Base Check (ohm-m)		Field Check (ohm-m)		
Shallow	46.0		46.0		
Deep	28.5		28.5		
Groningen	231.2		231.3		

Laterolog Calibration Tolerances MLE-D.A 283						
Shallow Resistor 2	1005.2	-3% 985.0 +3%	ohm			
Deep Resistor 2	1005.1	-3% 985.0 +3%	ohm			

Deep Resistor 2	1005.1		ohm
Groningen Resistor 2	1005.8		ohm
Shallow Base Check	46.0		ohm-m
Deep Base Check	28.5		ohm-m
Groningen Base Check	231.2		ohm-m
Shallow Field Check	46.0		ohm-m
Deep Field Check	28.5		ohm-m
Groningen Field Check	231.3		ohm-m

Laterolog Constants MLE-D.A 283		Last Edited on 04-SEP-2024,05:31	
Profiling Laterolog Type	Type 0		
Laterolog Output Filter	Logarithmic		
Profiling Limiter	On		
Median Filter	On		
Squasher Start	40000	ohm-m	
Shallow Laterolog K Factor	1.2844		
Deep Laterolog K Factor	0.7957		
Groningen Laterolog K Factor	0.8084		
Interference Rejection	50 Hz		
SP Connection	SP Bridle Electrode (Lower)		
Groningen Connection	Groningen Electrode (Upper)		
Profiling Deep Input	N/A		
Borehole Correction Constants			
Bridle Type	Standard		
Sonde Position	Centered	inches	
Hole Size Source	X Two Arm Caliper		
Hole Size Constant Value	N/A	inches	
Rm Source	Global Value: Temperature Corrected		
Temp. for Rm Corr.	MCG External Temperature		
Apparent Porosity and Water Saturation Constants			
Archie Constant (A)	1.00		
Cementation Exponent (M)	2.00		
Saturation Exponent (N)	2.00		
Saturation of Water for Apor	100.00	percent	
Resistivity of Water for Apor and Sw	0.05	ohm-m	
Resistivity of Mud Filtrate for Sw	0.00	ohm-m	
Source for Rt	0.00		
Source for Rxo	0.00		

Gamma Calibration MGS-D.A 252			Field Calibration on 04-SEP-2024 05:42
	Measured	Calibrated (API)	
Background	74	52	
Calibrator (Gross)	1092	759	
Calibrator (Net)	1017	707	

Gamma Calibration Tolerances MGS-D.A 252		
Ratio	1.439	Counts/API

Gamma Constants MGS-D.A 252			Last Edited on 04-SEP-2024,06:19
Gamma Calibrator Number	159		
GRC-M Calibrator Jig in Use?	NO		
Inactive Background Jig in Use?	NO		
Mud Density	1.10	gm/cc	
Caliper Source for Processing	X Two Arm Caliper		
Tool Position	Centred		
Potassium Equivalence	Chloride		
K Mud Concentration	0.00	%	

BEFORE SURVEY CALIBRATION

General Constants All 000

Last Edited on 04-SEP-2024,10:25

General Parameters		
Mud Resistivity	0.150	ohm-metres
Mud Resistivity Temperature	25.000	degrees C
Water Level	0.000	metres
Borehole Fluid Processing	Water Level Switch	
Hole/Annular Volume and Differential Caliper Parameters		
HVOL Method	Single Caliper	
HVOL Caliper 1	Density Caliper	
HVOL Caliper 2	N/A	
Annular Volume Diameter	5.500	inches
Caliper for Differential Caliper	None	
Rwa Parameters		
Porosity used	N/A	
Resistivity used	N/A	
RWA Constant A	N/A	
RWA Constant M	N/A	
SW/APOR Tool Source		

High Resolution Temperature Calibration MCG-E.A 531			Field Calibration on 03-SEP-2024,09:41
	Measured	Calibrated(Deg C)	
Lower	0.00	0.00	
Upper	100.00	100.00	

High Resolution Temperature Constants MCG-E.A 531		Last Edited on 03-SEP-2024,09:41
Pre-filter Length	11	

Caliper Calibration MMR-C.A 219			Base Calibration on 19-AUG-2024,08:15 Field Calibration on 04-SEP-2024 06:09
Base Calibration			
Reading No	Measured	Calibrator Size (in)	
1	15280	5.96	
2	18398	7.96	
3	21474	9.85	
4	25726	11.88	
5	0	0.00	
6	N/A	N/A	
Field Calibration			
	Measured Caliper (in)	Actual Caliper (in)	
	7.96	7.96	

Caliper Calibration Tolerances MMR-C.A 219		
Short Arm Field Cal.	7.96	7.76 7.96 8.16 in

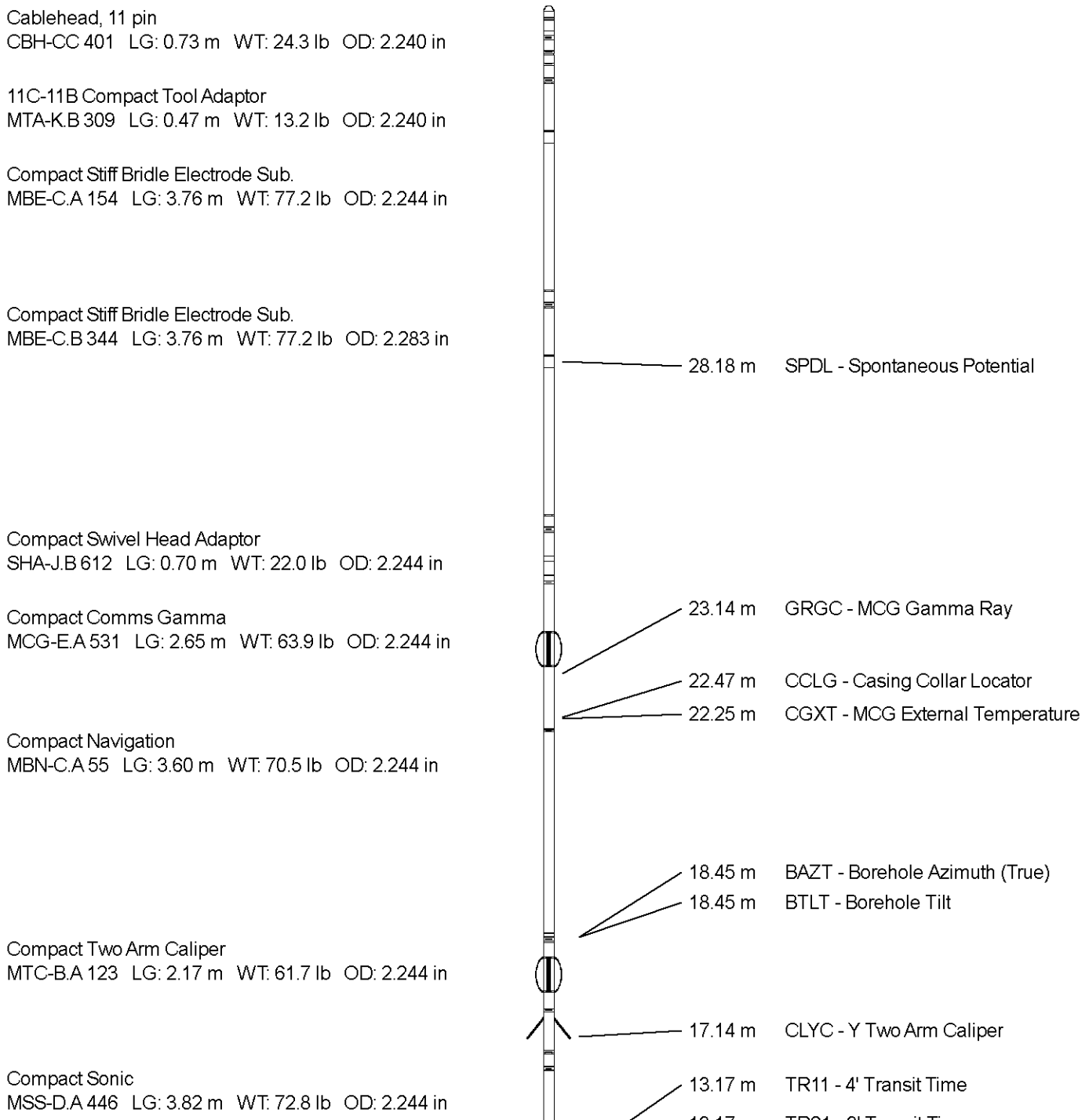
Micro Laterolog Calibration MMR-C.A 219				Base Calibration on 19-AUG-2024,08:16	
				Field Check on 04-SEP-2024 06:10	
Base Calibration	Resistor 1 (ohm)		Resistor 2 (ohm)		
	0.0		10000.0		
	Measured		Calibrated (ohm-m)		
	Ref 1	Ref 2	Ref 1	Ref 2	
	0.0	9989.2	0.0	128.0	
	Base Check (ohm-m)		Field Check (ohm-m)		
	5.1		5.1		

Micro Laterolog Calibration Tolerances MMR-C.A 219		
Ref 2	9989.2	-3% 9900.00 +3% ohm
Base Check	5.1	-2% 5.1 +2% ohm-m
		-2% 5.1 +2%

Pad Type	6 in Solid Nylon B23059	
Standoff Offset	0.0000	inches
Micro Laterolog K Factor	0.0128	
Micro Laterolog Rm K Factor	N/A	
Mudcake Thickness Correction Constants		
Mud Cake Source	Constant Value	
Mud Cake Thickness	0.1900	inches
Mud Cake Thickness Caliper	N/A	
Mud Cake Resistivity	0.1800	ohm-m
Mud Cake Resistivity Temp.	25.00	Degrees C
Mud Cake Resistivity Source	Temperature Corr	
Temp. for Rmc Corr.	MCG External Temperature	

DOWNHOLE EQUIPMENT

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Compact Two Arm Caliper
MTC-B.A 102 LG: 2.17 m WT: 61.7 lb OD: 2.240 in

Compact Knuckle Joint
SKJ-D 31 LG: 0.66 m WT: 24.3 lb OD: 2.244 in

Compact Upper Guard Sub.
MUG-B.A 139 LG: 2.74 m WT: 68.3 lb OD: 2.244 in

Compact Laterolog Electrode Sub.
MLE-D.A 283 LG: 3.76 m WT: 92.6 lb OD: 2.244 in

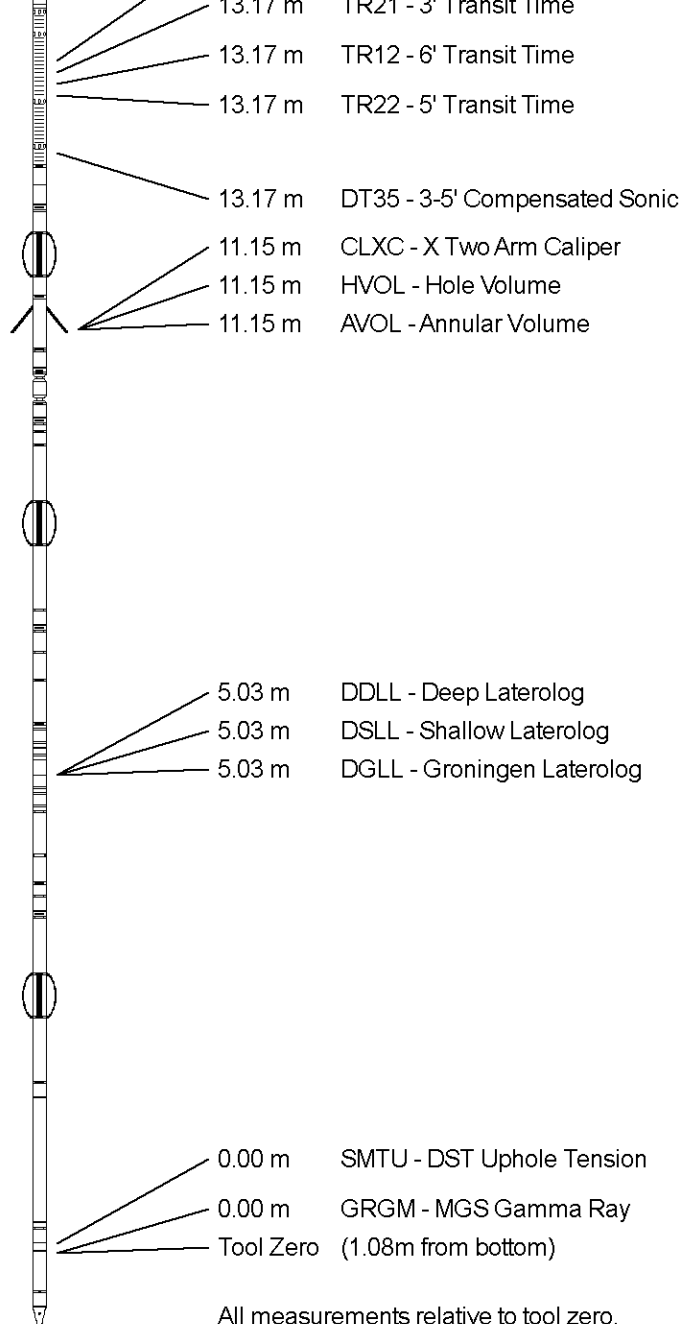
Compact Lower Guard Sub.
MLG-B.A 200 LG: 2.44 m WT: 55.1 lb OD: 2.244 in

Compact Tool Isolator sub.
MTI-B.A 74 LG: 0.47 m WT: 13.2 lb OD: 2.244 in

Compact Short Gamma
MGS-D.A 252 LG: 1.04 m WT: 24.3 lb OD: 2.244 in

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 2.244 in

Total Length: 35.20 m Weight: 828.9 lb



DOWNHOLE EQUIPMENT

C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView90\0\209_NADES_NUC_CALB.dta

Cablehead, 11 pin
CBH-CC 401 LG: 0.73 m WT: 24.3 lb OD: 2.240 in

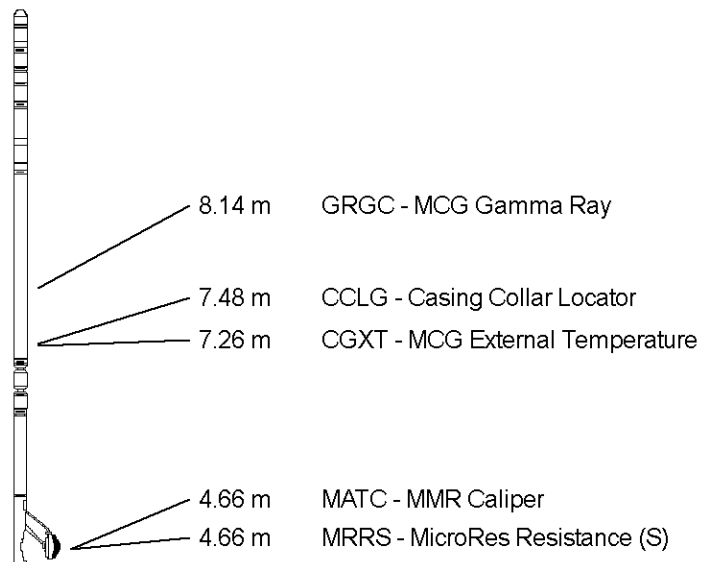
11C-11B Compact Tool Adaptor
MTA-K.B 309 LG: 0.47 m WT: 13.2 lb OD: 2.240 in

Compact Swivel Head Adaptor
SHA-J.B 612 LG: 0.70 m WT: 22.0 lb OD: 2.244 in

Compact Comms Gamma
MCG-E.A 531 LG: 2.65 m WT: 63.9 lb OD: 2.244 in

Compact Knuckle Joint
SKJ-E.A 255 LG: 0.66 m WT: 24.3 lb OD: 2.240 in

Compact Micro-Resistivity
MMR-C.A 219 LG: 2.62 m WT: 81.6 lb OD: 3.819 in

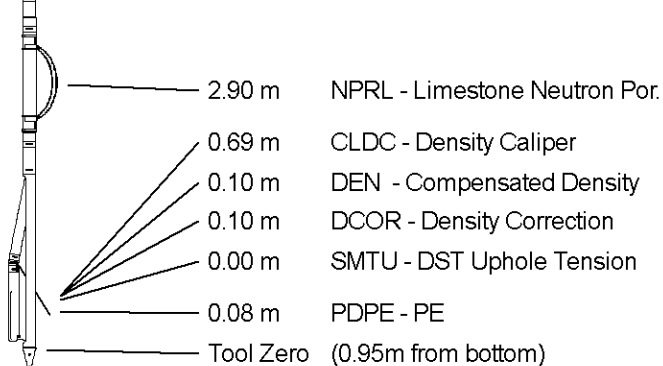


Compact Neutron
MDN-C.A 528 LG: 1.53 m WT: 50.7 lb OD: 2.244 in

Compact Density/Caliper
MPD-D.A 425 LG: 2.92 m WT: 90.4 lb OD: 2.449 in

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 2.244 in

Total Length: 12.56 m Weight: 377.0 lb



All measurements relative to tool zero.

COMPANY	S.N.G.N. ROMGAZ S.A.
WELL	209 NADES
FIELD	NADES
PROVINCE/COUNTY	MURES
COUNTRY/STATE	ROMANIA

Elevation Kelly Bushing	484.71	metres	Last Reading	250.00	metres
Elevation Drill Floor	484.71	metres	First Reading	1221.00	metres
Elevation Ground Level	480.21	metres	Depth Driller	1225.00	metres
			Depth Logger	1226.00	metres



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